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Systems, apparatus and methods for rotating a container in a container manipulation system

Abstract

Methods and apparatus for rotating a container delivered to a container manipulation system on a conveyor are described. In some embodiments, first and second carriages are positioned on opposite sides of the conveyor, each carrying respective forward and rear clamp arms. Each clamp arm is selectively independently moveable and carries at least first and second vertically spaced-apart wheels. All of the wheels are engageable with the container on the conveyor. At least two wheels on the first carriage are power-driven and selectively rotatable. The remaining wheels are free-spinning. Additional methods, systems and apparatus are also disclosed.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation application of and claims priority to U.S. patent application Ser. No. 17/492,644, filed on Oct. 3, 2021, entitled “Systems, Apparatus and Methods for Engaging a Cap Releasably Secured in a Container” and issued as U.S. Pat. No. 11,780,614 on Oct. 10, 2023, which is a continuation application of and claims priority to U.S. patent application Ser. No. 17/073,328 filed on Oct. 17, 2020, entitled “Apparatus for Installing a Bung Plug in and/or Extracting a Bung Plug from a Container” and issued as U.S. Pat. No. 11,136,150 on Oct. 5, 2021, which is a continuation application of and claims priority to U.S. patent application Ser. No. 16/190,647, filed on Nov. 14, 2018, entitled “Apparatus, Systems & Methods for Manipulating a Drum or other Container” and

which issued on Oct. 20, 2020 as U.S. Pat. No. 10,807,744, all of which are hereby incorporated by reference herein in their entireties.

FIELD OF THE DISCLOSURE

(1) The present disclosure relates generally to apparatus and methods for rotating containers in a container manipulation system.

BACKGROUND

(2) In various industries, containers (such as 55 gallon drums) are commonly used to store, transport and dispense materials and fluids. During the life of many such containers, after the containers have been initially filled or used, they are often refilled and reused many times. Other container processing activities may involve testing, cleaning, maintenance, retrofitting, etc. Automated or partially-automated systems and techniques have been developed for performing such container processing (“manipulation”) activities, which often include the removal and/or installation of the “bung plug” (or other closure) of the container.

(3) Presently known systems and techniques for manipulating containers are believed to have one or more disadvantages. For example, many known systems and techniques that require rotation of the container to move the container fill-hole to a desired position are ill-equipped to automatically, effectively and efficiently rotate the container, stop the container's rotation at a desired time, accurately position the fill hole or a combination thereof. For another example, many systems and techniques that involve removal and/or replacement of the bung plug are unable to effectively engage their bung plug extractor and/or installer with the container for extraction or installation when the container is not perfectly aligned therewith or has moved out of alignment. For yet another example, many known systems and techniques involving the replacement of threadably engaged bung plugs to various types of containers lack the ability to automatically, effectively, properly or efficiently thread the bung plugs or avoid frequent cross-threading. For still a further example, many such systems and techniques are not equipped to automatically, effectively, accurately or efficiently measure and/or report whether or not the bung plug has been properly installed in the container (e.g. whether an unacceptable or excessive amount of torque was applied to the installed bung plug or other variables), such as for Six Sigma, DOT or other requirements or certifications. With many known systems and techniques, it is believed to be difficult or impossible to perform one or more of the above operations (e.g. automatic, proper or effective container rotation, fill-hole positioning, bung plug extraction, bung plug installation, bung plug threading, measurement and reporting of bung plug installation variables) on a continuous supply of containers without frequent interruption, or with minimal or no human involvement. For still another example, various know systems and techniques are not equipped to automatically remove and install the same bung plug for each respective container. Further disadvantages are believed to exist with respect to various known systems or devices used for manipulating containers.

(4) It should be understood that the above-described examples, disadvantages, features and capabilities are provided for illustrative purposes only and are not intended to limit the scope or subject matter of this disclosure or the appended claims. Thus, none of the appended claims or claims of any patent or patent application related hereto should be limited by the above discussion or construed to address, include or exclude each or any of the above-cited examples, disadvantages, features and capabilities merely because of the mention thereof in this patent.

(5) Accordingly, there exists a need for improved systems, articles and methods useful for manipulating containers having one or more of the attributes or capabilities described or shown in, or as may be apparent from, the various parts of this patent.

BRIEF SUMMARY OF THE DISCLOSURE

(6) In various embodiments, the present disclosure involves apparatus for rotating a container in a container manipulation system. The container is delivered to the container manipulation system on a conveyor having first and second sides and front and rear ends. The apparatus includes first and

second carriages positioned proximate to opposite sides of the conveyor, respectively, so that the first carriage is closer to the first side of the conveyor and the second carriage is closer to the second side of the conveyor. Each carriage carries respective forward and rear clamp arms, the rear clamp arm of each carriage being closer to the rear end of the conveyor than the respective associated front clamp arm. Each clamp arm is selectively independently moveable toward and away from the container on the conveyor and carries at least first and second vertically spaced-apart wheels. The wheels on the respective rear clamp arms are rearward of the wheels on the corresponding associated front clamp arms relative to the conveyor. All of the wheels are engageable with the container on the conveyor. At least two of the wheels on the first carriage are power-driven and selectively rotatable. The remaining wheels on the first and second carriages are free-spinning.

(7) In some embodiment, the present disclosure involves apparatus for extracting a bung plug from and installing a bung plug into a container, the container having a bung opening formed in and facing upwardly from an upper surface thereof, the bung opening configured to contain the bung plug therein. A carrier is selectively movable up and down relative to the container. A bung plug engagement tool carried by the carrier includes at least one shaft having upper and lower ends and a bung plug wrench disposed at or proximate to the lower end of the at least one shaft. At least one vacuum source is associated with the bung plug engagement tool, whereby a vacuum may be selectively provided proximate to the bung plug wrench to retain the bung plug after it is extracted from the drum. A centering collar is carried by the carrier and at least partially encapsulates the bung plug wrench. The bung plug wrench and centering collar are both biased downwardly relative to the carrier and container and the bung plug wrench is additionally biased downwardly relative to the centering collar, carrier and container during use of the apparatus.

(8) If desired, the vacuum source may include at least one air compressor, vacuum generator or vacuum pump or a combination thereof. The vacuum source may be releasably fluidly coupled to the bung plug engagement tool. The centering collar may engage the container during bung plug installation and extraction.

(9) At least two spaced-apart outer springs may be positioned radially outwardly of the shaft(s) of the bung plug engagement tool, respectively, to provide downward biasing forces upon the centering collar and bung plug wrench relative to the carrier and container. The container may have a vertical axis that extends through the center of the bung opening and the carrier may have a vertical axis. The centering collar may engage the container during bung plug installation and extraction and the outer springs may allow the centering collar and bung plug wrench to concurrently tilt in any direction around 360° relative to the vertical axis of the carrier and automatically align in the vertical axis of the container during bung plug installation and extraction.

(10) At least two spaced-apart inner springs, separate and distinct from the outer springs, may also be positioned radially outwardly of the at least one shaft, respectively, and which provide downward biasing forces upon the bung plug wrench relative to the centering collar, carrier and container. The downward biasing forces of each outer spring may be greater than the downward biasing forces of each inner spring to retain the centering collar engaged with the container as desired during bung plug installation and extraction even if the bung plug wrench is pushed up. The inner springs may be configured to provide downward biasing forces on the bung plug wrench relative to the carrier, centering collar and container to assist in maintaining engagement of the bung plug wrench with the bung plug and container as desired during bung plug installation and extraction.

(11) In some embodiments, a method of extracting a bung plug from and installing a bung plug in a container is provided. The container has a bung opening formed in and facing upwardly from an upper surface thereof and the bung opening is configured to contain the bung plug therein. A carrier of the apparatus is selectively moved downwardly relative to the container and over the bung opening and bung plug contained therein. A centering collar carried by the carrier is downwardly

biased into contact with the upper surface of the container around the bung opening and a bung plug wrench carried by the carrier is downwardly biased into contact with the bung plug. The bung plug wrench is separately downwardly biased to ensure it stays in contact with the bung plug regardless of the position of the centering collar. At least one vacuum is associated with the bung plug wrench to retain the bung plug in contact with the bung plug wrench. The carrier bung plug carried thereby are selectively moved away from the container. The carrier is selectively moved downwardly relative to the container and over the bung opening. The centering collar is downwardly biased into contact with the upper surface of the container around the bung opening and the bung plug wrench is downwardly biased over the bung opening. The bung plug wrench aligns the bung plug with the bung opening. The vacuum associated with the bung plug wrench is released and the bung plug wrench engages the bung plug with the bung opening.

(12) In various embodiments, the present disclosure involves apparatus for extracting a bung plug from and installing a bung plug into a container having a bung opening formed in and facing upwardly from at least one surface thereof. The bung opening is configured to contain the bung plug therein. The apparatus includes a carrier selectively movable up and down relative to the container. A bung plug engagement tool is carried by the carrier and includes at least one shaft a bung plug wrench disposed at or proximate to the lower end thereof. A centering collar is also carried by the carrier. During use of the apparatus, the bung plug wrench and centering collar are both biased downwardly relative to the carrier and container and the bung plug wrench is additionally biased downwardly relative to the centering collar, carrier and container. At least two spaced-apart outer springs are positioned radially outwardly of the shaft(s) of the bung plug engagement tool, respectively, and provide downward biasing forces upon the centering collar and bung plug wrench relative to the carrier and container.

(13) If desired, at least two spaced-apart inner springs, separate and distinct from the outer springs, may also be positioned radially outwardly of the shaft(s) and provide downward biasing forces upon the bung plug wrench relative to the centering collar, carrier and container. The centering collar may engage at least one surface of the container during bung plug installation and extraction and the downward biasing forces of each outer spring may be greater than the downward biasing forces of each inner spring. The springs may take the form of resilient members, elastic members, coil springs, swivel joint, bearings, non-resilient members, a motor or any other form.

(14) The container may have a vertical axis that extends through the center of the bung opening and the carrier may have a vertical axis. During bung plug installation and extraction, the centering collar may engage the container and the outer springs may allow the centering collar and bung plug wrench to concurrently tilt in any direction around 360° relative to the vertical axis of the carrier and automatically align in the vertical axis of the container.

(15) In many embodiments, the present disclosure involves apparatus for extracting a bung plug from and installing a bung plug into a container. The container has a bung opening formed in at least one surface thereof and which is configured to contain the bung plug therein. The apparatus includes a carrier selectively movable up and down relative to the container. A bung plug engagement tool is carried by the carrier and includes at least one shaft and a bung plug wrench disposed at or proximate to the lower end thereof. A centering collar is carried by the carrier. Both the bung plug wrench and centering collar are biased downwardly relative to the carrier and container during use of the apparatus. The bung plug wrench is additionally biased downwardly relative to the centering collar, carrier and container. At least two spaced-apart inner springs provide downward biasing forces upon the bung plug wrench relative to the centering collar, carrier and container. If desired, the inner springs may be configured to assist in maintaining engagement of the bung plug wrench with the bung plug as desired during bung plug installation and extraction.

(16) Certain embodiments involve methods of engaging a bung plug extractor/installer with a bung plug that is releasably secured to a container. The container includes a bung opening formed in and

facing upwardly from at least one surface thereof and which is configured to contain the bung plug therein. The method includes selectively moving a carrier of the bung plug extractor/installer over the bung opening and bung plug contained therein. The carrier carries a centering collar and bung plug engagement tool, the bung plug engagement tool including at least one shaft and at least one bung plug wrench. At least first and second spaced-apart springs are positioned radially outwardly of at least one shaft of the bung plug engagement tool and assist in biasing the centering collar downwardly into contact with at least one surface of the container around the bung opening and biasing at least one bung plug wrench downwardly into contact with the bung plug. At least third and fourth spaced-apart springs separately downwardly bias the bung plug wrench relative to the centering collar, carrier and container.

(17) If desired, these methods may include actuating the bung plug wrench to disconnect the bung plug from the container. Suction pressure may be provided in at least one passageway formed in at least part of the shaft(s) of the bung plug engagement tool to retain the bung plug in contact with the bung plug wrench. The carrier and bung plug carried thereby may be selectively moved away from and/or over the bung opening of the container. A centering assembly carried by the carrier may align the bung plug with the bung opening. The centering collar may be biased into contact with at least one surface of the container around the bung opening and the bung plug may be biased onto the bung opening. Suction pressure in the at least one passageway may be released and the bung plug secured in the bung opening.

(18) There are embodiments of the present disclosure that involves apparatus for extracting a cap from a fill hole of a container with the use of at least one vacuum source that provides suction pressure. The fill hole is formed in at least one surface of the container. The apparatus includes a carrier selectively movable up and down relative to the container. A cap engagement tool is carried by the carrier and selectively at least partially engageable with the cap. When the cap engagement tool engages the cap, at least one space is formed between at least part of the cap engagement tool and the cap. A centering collar is carried by the carrier and positioned at least partially radially outwardly of at least part of the cap engagement tool. A first plurality of spaced-apart biasing members is positioned radially outwardly of at least part of the cap engagement tool and provides downward biasing forces upon the centering collar and at least part of the cap engagement tool relative to the carrier and container. At least one passageway fluidly couples one or more of the spaces and vacuum source and allows suction pressure to be selectively provided to the space(s) to retain the cap with the cap engagement tool after the cap is extracted from the container.

(19) If desired, the cap engagement tool may include at least one shaft and a cap extraction wrench disposed at or proximate to the lower end thereof and the fluid passageway(s) may extend at least partially through the shaft(s) and cap extraction wrench. A second plurality of spaced-apart biasing members, separate and distinct from the first plurality of spaced-apart biasing members, may be provided to provide downward biasing forces upon at least part of the cap engagement tool relative to the centering collar, carrier and container. The first and/or second pluralities of spaced-apart biasing members may include three spaced-apart biasing members. If desired, each biasing member may include at least one among at least one coil spring, resilient member, non-resilient member, swivel and bearing.

(20) The present disclosure also includes embodiments of apparatus for extracting a cap from a fill hole of a container with the use of at least one vacuum source that provides suction pressure. The fill hole is formed in at least one surface of the container. The apparatus includes a carrier selectively movable up and down relative to the container. A cap engagement tool is carried by the carrier and selectively at least partially engageable with the cap. When the cap engagement tool engages the cap, at least one space is formed between at least part of the cap engagement tool and the cap. A centering collar is carried by the carrier and positioned at least partially radially outwardly of at least part of the cap engagement tool. A first plurality of spaced-apart biasing members provides downward biasing forces upon at least part of the cap engagement tool relative

to the centering collar, carrier and container. At least one passageway fluidly couples one or more of the spaces and vacuum source and allows suction pressure to be selectively provided to the space(s) to retain the cap with the cap engagement tool after the cap is extracted from the container. (21) If desired, a second plurality of spaced-apart biasing members which are separate and distinct from the first plurality of spaced-apart biasing members may provide downward biasing forces upon the centering collar relative to the carrier and container. The downward biasing forces of each of the second plurality of biasing members may be greater than the downward biasing forces of each of the first plurality of biasing members. If desired, each biasing member may include at least one among at least one coil spring, resilient member, non-resilient member, swivel and bearing. (22) In some embodiments, the present disclosure involves apparatus for installing a removable bung plug in and/or extracting a removable bung plug from a container positioned below the apparatus. The container has a bung opening formed in and facing upwardly from an upper surface thereof. The bung opening is configured to contain the bung plug therein. The apparatus includes a carrier selectively movable up and down relative to the container. The carrier includes a base plate having a central bore with (i) a conically-shaped inner surface and (ii) minimum and maximum inner diameters. A downwardly-biased, power-driven bung plug engagement tool is carried by the carrier and has a bung plug wrench at the lower end thereof. A downwardly-biased centering assembly is also carried by the carrier and has (i) a conically-shaped outer surface extending at least partially therearound closer to the upper end than the lower end thereof and (ii) a centering collar located proximate to the lower end thereof. The centering collar extends at least partially around the bung plug wrench and is concurrently moveable laterally and angularly therewith relative to the carrier and container. The conically-shaped outer surface of the centering assembly is shaped and sized to seat within the bore of the base plate and be moveable up and then down therein and relative to the base plate, carrier and container. The width or outer diameter of the centering assembly immediately below the conically-shaped outer surface is smaller than the minimum inner diameter of the bore of the base plate so that after the conically-shaped outer surface of the centering assembly moves up relative to the bore of the base plate, the centering assembly and bung plug wrench are concurrently movable laterally relative to the carrier and container.

(23) In various embodiments, the present disclosure involves an apparatus for installing a bung plug in and/or extracting a bung plug from a container positioned below the apparatus. The container includes a bung opening formed in and facing upwardly from an upper surface thereof. The bung opening is configured to contain the bung plug therein. The apparatus include a carrier selectively movable up and down relative to the container. A power-driven bung plug engagement tool is carried by the carrier and includes (i) an elongated, vertically-oriented shaft having upper and lower ends and (ii) a bung plug wrench disposed at the lower end of the shaft and configured to engage the bung plug in the bung opening of the container during installation of the bung plug and/or disengage the bung plug from the bung opening during extraction of the bung plug. A centering collar is also carried by the carrier and at least partially encapsulates the bung plug wrench. At least two spaced-apart outer springs are positioned radially outwardly of the shaft, respectively, and provide downward biasing forces upon the centering collar and bung plug wrench relative to the carrier and container. At least two spaced-apart inner springs, separate and distinct from the outer springs, are also positioned radially outwardly of the shaft, respectively, and provide downward biasing forces upon the bung plug wrench relative to the centering collar, carrier and container.

(24) The present disclosure includes embodiments of apparatus for installing a bung plug in and/or extracting a bung plug from a container positioned below the apparatus. The container has a bung opening formed in and facing upwardly from an upper surface thereof and an upwardly protruding bung opening lip extending at least partially around the bung opening. The bung opening lip has an outer diameter and the bung opening is configured to contain the bung plug therein. The apparatus

includes a carrier selectively movable up and down relative to the container and including a base plate having a central bore with a conically-shaped inner surface. A downwardly-biased, power-driven bung plug engagement tool is carried by the carrier and includes a bung plug wrench proximate to the lower end thereof. A downwardly-biased centering collar is also carried by the carrier, has a lowermost edge and an inner diameter. The centering collar extends at least partially around the bung plug wrench and is concurrently moveable laterally and angularly therewith relative to the carrier and container. A downwardly-biased centering cone is coupled to the centering collar and has a conically-shaped outer surface extending at least partially around the perimeter thereof and which is shaped and sized to seat within the bore of the base plate and be moveable up and then down therein and relative to the base plate and the carrier. The conically-shaped outer surface of the centering cone is spaced upwardly from the lowermost edge of the centering collar no greater than twice the smaller of the outer diameter of the bung opening lip or inner diameter of the centering collar.

(25) There are also embodiments herein of apparatus for installing a removable, threaded bung plug in a container positioned below the apparatus. The container has a vertical axis, an upper surface and a threaded bung opening formed in and facing upwardly from the upper surface. The vertical axis of the container extends through the center of the bung opening. The threads of the bung opening are perpendicular to the vertical axis of the container and configured to engage threads of the bung plug. The apparatus includes a carrier selectively movable up and down relative to the container. A downwardly-biased, power-driven, rotatable bung plug engagement tool is carried by the carrier and includes a bung plug wrench at the lower end thereof. The bung plug wrench is configured to threadably engage the bung plug in the bung opening of the container. A positionally self-adjusting, downwardly-biased centering assembly is also carried by the carrier and includes a centering collar located proximate to the lower end thereof. The centering collar extends at least partially around the bung plug wrench and is concurrently moveable laterally and angularly with the bung plug wrench relative to the carrier and container. As the carrier is moved down to the container, the centering assembly will automatically align the centering collar and bung plug wrench in the vertical axis of the container and perpendicular to the threads of the bung opening. At least one displacement sensor is coupled to the centering assembly and moveable concurrently therewith so that as the carrier is moved down to the container, the centering assembly will automatically position the displacement sensor(s) to accurately measure the amount of vertical movement of the bung plug in the bung opening during installation thereof.

(26) In many embodiments, the present disclosure involves apparatus for installing a removable bung plug in a container positioned below the apparatus. The container has a bung opening formed in and facing upwardly from an upper surface thereof and an upwardly protruding bung opening lip extending at least partially around the bung opening. The bung opening is configured to contain the bung plug therein. The apparatus includes a carrier selectively movable up and down relative to the container. A downwardly-biased bung plug engagement tool is carried by the carrier and has a bung plug wrench at the lower end thereof. A downwardly-biased centering assembly is also carried by the carrier and includes a centering collar located proximate to the lower end thereof. The centering collar extends at least partially around the bung plug wrench and is positioned to sit down upon the bung opening lip during bung plug installation without directly abutting the upper surface of the container. At least one displacement sensor is rigidly coupled to the centering assembly so that as the carrier is moved down to the container, the displacement sensor(s) will be positioned to accurately measure the depth of the bung plug in the bung opening throughout installation thereof on a continuing basis in real time.

(27) In some embodiments, the present disclosure includes apparatus for extracting a removable bung plug from a container positioned below the apparatus. The container has a bung opening formed in and facing upwardly from an upper surface thereof and an upwardly protruding bung opening lip extending at least partially around the bung opening. The bung opening is configured to

contain the bung plug therein. The apparatus includes a carrier selectively movable up and down relative to the container. A downwardly-biased, power-driven bung plug engagement tool is carried by the carrier and includes (i) an elongated, vertically-oriented shaft having upper and lower ends and (ii) a bung plug wrench disposed at the lower end of the shaft. After the carrier is moved down to the container, the bung plug engagement tool is configured to engage and remove the bung plug from the container and thereafter carry the bung plug as the carrier is moved up away from the container. A bung plug engagement tool retractor is operatively coupled to the bung plug engagement tool. A downwardly-biased centering assembly is carried by the carrier includes a centering collar located proximate to the lower end thereof and extending at least partially around the bung plug wrench. After the bung plug engagement tool removes the bung plug from the container and before the carrier is moved up away from the container, the bung plug engagement tool retractor is configured to move the bung plug engagement tool and bung plug upwardly relative to the centering assembly, carrier and container and clear of the bung opening lip.

(28) In various embodiments, apparatus for installing a threaded bung plug in a container positioned below the apparatus includes a carrier selectively movable up and down relative to the container. The carrier includes a base plate having a central bore with a conically-shaped inner surface. A power-driven bung plug engagement tool is carried by the carrier and includes a rotatable bung plug wrench at the lower end thereof. A centering assembly is carried by the carrier and includes a conically-shaped outer surface extending at least partially therearound closer to the upper end than the lower end thereof. The conically-shaped outer surface of the centering assembly is shaped and sized to seat within the bore of the base plate and is moveable up and then down therein and relative to the base plate, carrier and container. The centering assembly also includes a centering collar located proximate to the lower end thereof. The centering collar extends at least partially around the bung plug wrench and is concurrently moveable side-to-side and angularly therewith relative to the carrier and container. Whenever the conically-shaped outer surface of the centering assembly moves up relative to the bore of the base plate during bung plug installation, the centering assembly and bung plug wrench are concurrently movable side-to-side and angularly relative to the carrier and container. At least one automated torque sensor is operatively coupled to the bung plug wrench and configured to measure torque placed upon, and count the number of revolutions of rotation of, the bung plug during installation thereof in real time.

(29) In some embodiments, apparatus for installing a threaded bung plug in a container positioned below the apparatus includes a carrier selectively movable up and down relative to the container. The carrier includes a base plate having a central bore with a conically-shaped inner surface. A power-driven bung plug engagement tool is carried by the carrier and includes a bung plug wrench. A centering collar is carried by the carrier and extends at least partially around the bung plug wrench. A centering cone has a conically-shaped outer surface extending at least partially around the perimeter thereof, is shaped and sized to seat within the bore of the base plate and moveable up and then down therein and relative to the base plate and the carrier. At least one displacement sensor is arranged to accurately measure the amount of vertical movement of the bung plug in the bung opening during installation thereof. At least one electronic controller is electronically coupled to the displacement sensor(s) to receive measurements taken thereby in real-time during installation of the bung plug and, based at least partially upon at least one such measurement, determines if the bung plug has been cross-threaded in the threaded bung opening and, if so, stops installation of the bung plug.

(30) In many embodiments, apparatus for installing a removable, threaded bung plug in a container positioned below the apparatus and having a vertical axis, an upper surface and a threaded bung opening formed in and facing upwardly from the upper surface. The vertical axis of the container extends through the center of the bung opening. The threads of the bung opening are perpendicular to the vertical axis of the container and configured to engage threads of the bung plug. The apparatus includes a carrier selectively movable up and down relative to the container. A rotatable

bung plug engagement tool is carried by the carrier and includes a bung plug wrench at the lower end thereof. The bung plug wrench is configured to threadably engage the bung plug in the bung opening of the container. A positionally self-adjusting centering assembly carried by the carrier has a centering collar located proximate to the lower end thereof. The centering collar extends at least partially around the bung plug wrench and is concurrently moveable side-to-side and angularly therewith relative to the carrier and container. As the carrier is moved down to the container, the centering assembly will automatically align the centering collar and bung plug wrench in the vertical axis of the container and perpendicular to the threads of the bung opening. At least one linear encoder is coupled to the centering assembly and moveable concurrently therewith, whereby as the carrier is moved down to the container, the centering assembly will automatically position the at least one linear encoder to accurately measure the amount of vertical movement of the bung plug in the bung opening during installation thereof.

(31) In certain embodiments, apparatus for installing a removable bung plug in a container positioned below the apparatus includes a carrier selectively movable up and down relative to the container. A bung plug engagement tool is carried by the carrier and includes a bung plug wrench. A centering assembly is also carried by the carrier and includes a centering collar located proximate to the lower end thereof. The centering collar extends at least partially around the bung plug wrench and is positioned to at least partially sit down upon the bung opening lip during bung plug installation. At least one displacement sensor is configured to measure the depth of the bung plug in the bung opening during installation thereof. At least one torque sensor is operatively coupled to the bung plug wrench and configured to measure torque placed upon the bung plug as the bung plug wrench installs the bung plug in the bung opening and count the number of revolutions of rotation of the bung plug as it is installed in the container.

(32) This disclosure includes apparatus for installing a bung plug in a drum positioned below the apparatus. The drum has a bung opening formed in and facing upwardly from an upper surface thereof. The bung opening is configured to contain the bung plug therein. The apparatus includes a carrier selectively movable up and down relative to the drum. A bung plug engagement tool carried by the carrier includes at least one shaft a bung plug wrench disposed proximate to the lower end of the shaft. A centering collar is also carried by the carrier and at least partially encapsulates the bung plug wrench. At least two spaced-apart outer biasing members are positioned radially outwardly of the shaft and provide downward biasing forces upon the centering collar and bung plug wrench relative to the carrier and drum. At least two spaced-apart inner biasing members, separate and distinct from the outer biasing members, are also positioned radially outwardly of the shaft and provide downward biasing forces upon the bung plug wrench relative to the centering collar, carrier and drum.

(33) In many embodiments, apparatus for extracting a bung plug from a drum positioned below the apparatus can be used with a drum having a bung opening formed in and facing upwardly from an upper surface thereof. The bung opening is configured to contain the bung plug therein. The apparatus includes a carrier selectively movable up and down relative to the drum. A bung plug engagement tool is carried by the carrier and including at least one shaft having a bung plug wrench disposed proximate to the lower end thereof. A centering collar is also carried by the carrier and at least partially encapsulates the bung plug wrench. At least two spaced-apart outer biasing members are positioned radially outwardly of the shaft, respectively, and provide downward biasing forces upon the centering collar and bung plug wrench relative to the carrier and drum. At least two spaced-apart inner biasing members, separate and distinct from the outer biasing members, are also positioned radially outwardly of the shaft, and provide downward biasing forces upon the bung plug wrench relative to the centering collar, carrier and drum.

(34) In many embodiments, the present disclosure involves apparatus for installing a removable, threaded bung plug in a container positioned below the apparatus. The container has a vertical axis, an upper surface and a threaded bung opening formed in and facing upwardly from the upper

surface. The vertical axis of the container extends through the center of the bung opening, while the threads of the bung opening are perpendicular to the vertical axis of the container and configured to engage threads of the bung plug. The apparatus includes a carrier selectively movable up and down relative to the container. A bung plug engagement tool is carried by the carrier and includes a rotatable bung plug wrench. A positionally self-adjusting centering assembly is also carried by the carrier and includes centering collar located proximate to the lower end thereof. The centering collar extends at least partially around the bung plug wrench. At least one displacement sensor is arranged to measure the amount of vertical movement of the bung plug in the bung opening during installation thereof. At least one torque sensor is associated with the bung plug wrench and arranged to measure torque placed upon, and count the number of revolutions of rotation of, the bung plug during installation thereof.

(35) The present disclosure includes embodiments of an automated system for removing a bung plug from a drum and installing a bung plug in the drum. The drum is movable on a conveyor to and from at least first and second stations associated with the system. The first and second stations may or may not be immediately adjacent to one other and there may be one or more intermediate stations between the first and second stations. The drum includes a bung opening formed in and facing upwardly from an upper surface thereof and an outer rim extending around the upper surface. The bung opening is configured to contain a removable bung plug therein. The center of the bung opening is spaced inwardly from the outer rim of the drum by a distance. The bung opening has an upwardly protruding lip extending therearound and a diameter measured between opposing sides of the lip.

(36) The automated system of these embodiments includes an automated drum rotator and an automated, power-driven, up-and-down moving bung locator at the first station. The bung locator has at least first and second downwardly-facing bung location sensors spaced apart the same distance as the diameter of the bung opening. The bung locator also carries at least one downwardly-extending foot spaced outwardly relative to the bung location sensors and configured to rest upon the outer rim of the drum. The bung locator is configured to be lowered down over the drum until at least one foot thereof contacts the outer rim of the drum, whereby the positioning of at least one foot of the bung locator in contact with outer rim will position each of the bung location sensors radially inwardly of the outer rim the same distance as the distance between the outer rim and the center of the bung opening. As the drum is rotated below the bung locator, the bung location sensors will concurrently detect the presence of opposing sides of the lip of the bung opening thereunder, respectively.

(37) The automated system of these embodiments also includes an automated, power-driven, downwardly-facing bung plug extractor at the first station. The bung plug extractor includes an automated, power-driven, bung plug wrench and a centering collar extending at least partially around the bung plug wrench. The bung plug extractor further includes at least two outer springs providing downward biasing forces upon the centering collar thereof and positioned radially outwardly of and above the centering collar. When there is a bung plug in the bung opening of the drum at the first station and the bung plug extractor is aligned with and lowered over the bung opening, the centering collar thereof will rest at least partially upon the lip of the bung opening and the bung plug wrench thereof will be positioned to engage and loosen the bung plug from the drum. If the bung plug extractor and bung opening are not aligned as the bung plug extractor is lowered, one or more of the outer springs of the bung plug extractor may deflect to allow the centering collar and bung plug wrench of the bung plug extractor to tilt relative to the drum to align the bung plug wrench over the bung plug and allow its engagement therewith.

(38) The automated system of these embodiments also includes an automated, power-driven, downwardly-facing bung plug installer at the second station. The bung plug installer is distinct from the bung plug extractor and includes an automated, power-driven, bung plug wrench and a centering collar extending at least partially around the bung plug wrench. The bung plug installer

further includes at least two outer springs providing downward biasing forces upon the centering collar thereof and positioned radially outwardly of and above the centering collar. When the bung plug installer is aligned and lowered over the bung opening of the drum, the centering collar thereof will rest at least partially upon the lip of the bung opening and the bung plug wrench thereof will be radially inwards thereof and selectively power-driven to install a bung plug in the bung opening. If the bung plug installer and bung opening are not aligned as the bung plug installer is lowered, one or more of the outer springs of the bung plug installer may deflect to allow the centering collar and bung plug wrench of the bung plug installer to tilt relative to the drum to align the bung plug wrench over the bung opening for installation of a bung plug therein.

(39) If desired, the automated drum rotator may include front and rear wheeled carriages positioned on opposing sides of the conveyor at the first station. Each carriage may include at least a first automated, power-driven clamp arm, each clamp arm having at least one rotatable wheel mounted thereon. Each clamp arm may be independently selectively moveable inwardly relative to the conveyor and the drum positioned at the first station until at least one wheel thereof contacts the drum. At least one of the wheels of the front and/or rear wheeled carriages may be automated and power-driven to selectively rotate the drum. Each carriage may include automated outer and inner, independently moving, power-driven clamp arms, respectively. Each clamp arm may include at least two spaced-apart wheels thereon. Each wheel may be engageable with the drum. Some or all of the wheels of the front wheeled carriage may be power-driven to selectively rotate the drum.

(40) If desired, the system may include a rotation platform positioned under the drum at the first station and selectively moveable up and down relative to the conveyor. The rotation platform can be lifted with the drum on it during rotation of the drum. The bung location sensors may be fiber optics sensors. At least one optic amplifier may be associated with the bung location sensors and configured to cause one or more light signals to project down through each bung location sensor onto the upper surface of the rotating drum, receive light signals reflected up off the upper surface the drum through the bung location sensors and determine when reflected light signals are concurrently received through both bung location sensors to indicate the bung location sensors are simultaneously positioned over respective opposing edges of the lip of the bung opening.

(41) The bung locator may not engage or ride across the upper surface of the drum. The foot or feet of the bung locator may be rigidly coupled to the bung locator and not include any moving parts. The bung plug extractor and/or bung plug installer may include three outer springs evenly spaced-apart around the centering collar thereof and configured to allow the centering collar and bung plug wrench thereof to concurrently tilt in any direction around 360° relative to the drum. The bung plug extractor and/or installer may be configured to allow up to ¼" displacement of the centering collar and bung plug wrench thereof in any direction around 360° relative to the drum.

(42) The system may include an electronic controller electronically coupled to the bung locator, bung plug extractor and bung plug installer and configured to automatically control the movement thereof, respectively. The bung locator may be configured to emit a series of reflective signals through the bung location sensors onto the upper surface of the drum, retrieve at least some of the reflective signals and communicate data relating thereto to the electronic controller. The system may include an air motor electronically coupled to the electronic controller and operatively coupled to the bung plug wrench of the bung plug extractor and configured to selectively rotate the bung plug wrench and bung plug to automatically loosen the bung plug from the drum. A servo-motor may be electronically coupled to the electronic controller and operatively coupled to the bung plug wrench of the bung plug installer and configured to selectively rotate the bung plug wrench and bung plug to automatically install the bung plug in the bung opening of the drum.

(43) If desired, a vacuum source may be associated with the system. The bung plug wrench of the bung plug extractor and/or installer may include upper and lower ends and at least one vacuum space proximate to its lower end. The vacuum space of the bung plug wrench of the bung plug extractor may be fluidly coupled to the vacuum source to provide suction in the vacuum space so

the bung plug wrench of the bung plug extractor can be used to effectively lift up the bung plug after it is removed from the drum and effectively carry it to a bung plug depository and selectively release suction in the vacuum space when the bung plug wrench and bung plug carried thereby are positioned over the bung plug depository. The vacuum space of the bung plug wrench of the bung plug installer may be fluidly coupled to the vacuum source to provide suction in the vacuum space of the bung plug installer so the bung plug wrench of the bung plug installer can be used to effectively lift a bung plug from the bung plug depository and effectively carry it to the bung opening of the drum and install it therein.

(44) The present disclosure also includes embodiments of an automated system for installing a bung plug in a drum. The drum is movable on a conveyor into and through the system. The drum includes a bung opening formed in and facing upwardly from an upper surface thereof and an outer rim extending around the upper surface. The bung opening is configured to contain a releasable bung plug threadably engageable therein and the bung opening has an upwardly protruding lip extending therearound. The system includes an electronic controller and automated drum rotator electronically coupled to the electronic controller. The electronic controller controls use of the drum rotator to selectively rotate the drum in the system. An automated, power-driven, up-and-down moving bung locator is electronically coupled to the controller and carries at least one downwardly-facing bung location sensor. The bung locator includes at least one downwardly-extending foot spaced outwardly relative to the bung location sensor(s) and configured to rest at least partially upon the outer rim of the drum. The bung locator is configured to be lowered down over the drum until at least one foot thereof contacts the outer rim of the drum. As the drum is rotated below the bung locator, the bung location sensor(s) will detect the presence of the lip of the bung opening thereunder.

(45) In these embodiments, an automated, power-driven, downwardly-facing bung plug installer is also electronically coupled to the electronic controller. The bung plug installer includes an automated, power-driven, bung plug wrench selectively rotatable to install a bung plug in the bung opening. An automated torque sensor is operatively coupled to the bung plug wrench of the bung plug installer and electronically coupled to the electronic controller. The torque sensor is configured to measure torque placed upon the bung plug as the bung plug wrench of the bung plug installer installs the bung plug in the bung opening in real-time as often as desired. An automated displacement sensor is coupled to the bung plug installer and electronically coupled to the electronic controller. The displacement sensor is configured to measure the amount of vertical movement of the bung plug in the bung opening during installation thereof. Based at least partially upon at least one measurement taken by the torque sensor and/or the displacement sensor, the electronic controller will stop the installation of the bung plug by the bung plug wrench of the bung plug installer and/or determine if the amount of torque applied to the bung plug meets, exceeds or falls below a desired torque value.

(46) If desired, the displacement sensor may include at least one linear encoder. The bung plug installer may further include a servo-motor electronically coupled to the electronic controller and operatively coupled to the bung plug wrench of the bung plug installer and configured to selectively rotate the bung plug wrench and bung plug to install the bung plug in the bung opening of the drum. The automated torque sensor may include at least one encoder disposed within the servo-motor. The automated torque sensor may be configured to count the number of revolutions of rotation of the bung plug as it is installed in the drum.

(47) In some embodiments, the present disclosure involves an automated method of removing a bung plug from a drum and installing a bung plug in the drum with the use of an automated drum manipulation system. The drum has a bung opening formed in and facing upwardly from its upper surface and an outer rim extending around the upper surface. The bung opening is configured to contain a bung plug therein. The center of the bung opening is spaced inwardly from the outer rim of the drum by a distance. The bung opening has an upwardly protruding lip extending

therearound.

(48) In these embodiments, after the drum with a bung plug in its bung opening arrives in the system, at least one automated, power-driven, clamp arm of a front and/or a rear wheeled carriage moves inwardly toward the drum until at least one wheel thereof contacts the drum. At least two automated, power-driven wheels of at least one of the carriages rotate the drum. An automated bung locator moves down to position at least one foot extending downwardly therefrom upon the outer rim of the drum to automatically position first and second downwardly-facing, spaced-apart bung location sensors carried by the bung locator radially inwardly of the outer rim of the drum the same approximate distance as the distance between the outer rim and the center of the bung opening. As the drum is rotated below the bung locator, the first and second bung location sensors automatically detect when opposing sides of the lip of the bung opening are concurrently positioned thereunder. The power-driven wheel(s) of the front and/or rear carriages automatically cease rotating the drum when or after the first and second bung location sensors detect opposing sides of the lip of the bung opening are concurrently positioned thereunder.

(49) An automated downwardly-facing bung plug extractor moves down over the bung hole to position a centering collar thereof atop the lip of the bung opening and an automated, power-driven bung plug wrench of the bung plug extractor radially inwards of the lip. The bung plug wrench loosens the bung plug from the drum and the bung plug is removed from the drum. An automated downwardly-facing bung plug installer (separate and distinct from the bung plug extractor) moves down over the bung hole of the drum to position a centering collar thereof atop the lip of the bung opening and an automated, power-driven bung plug wrench of the bung plug installer radially inwards of the lip. The bung plug wrench of the bung plug installer installs a bung plug in the bung opening of the drum. If desired, during installation of the bung plug in the bung opening, an automated torque sensor operatively coupled to the bung plug wrench of the bung plug installer may measure torque placed upon the bung plug by the bung plug wrench of the bung plug installer during installation thereof in real-time. Based at least partially upon at least one measurement taken by the automated torque sensor, an electronic controller electronically coupled to the automated torque sensor may determine if the amount of torque applied to the bung plug meets, exceeds or falls below a desired torque value.

(50) Accordingly, the present disclosure includes features and advantages which are believed to enable it to advance container manipulation technology. Characteristics and advantages of the present disclosure described above and additional features and benefits will be readily apparent to those skilled in the art upon consideration of the following detailed description of various embodiments and referring to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The following figures are part of the present specification, included to demonstrate certain aspects of various embodiments of this disclosure and referenced in the detailed description herein:

(2) FIG. 1A is a perspective view of an embodiment of a drum manipulation system shown with an exemplary drum filling system in accordance with the present disclosure;

(3) FIG. 1B is a perspective view of various exemplary components of the drum manipulation system of FIG. 1A;

(4) FIG. 1C is an exploded view of various exemplary components of the drum manipulation system of FIG. 1A including the bung locator and bung plug extractor in accordance with one or more embodiments of the present disclosure;

(5) FIG. 2 is a perspective view of the exemplary drum manipulation system and drum filling system of FIG. 1A along with an exemplary framework and enclosure in accordance with one or

more embodiments of the present disclosure;

(6) FIG. 3 is a perspective view of an exemplary steel drum;

(7) FIG. 4 is a perspective view of an exemplary metallic bung plug;

(8) FIG. 5 is a perspective view of an exemplary plastic bung plug;

(9) FIG. 6 is an exploded view of various components of the exemplary drum manipulation system of FIG. 1A including the exemplary drum rotator in accordance with one or more embodiments of the present disclosure;

(10) FIG. 7 is a perspective view of various components of the exemplary drum manipulation system of FIG. 1A including the exemplary bung locator and bung plug extractor in accordance with the present disclosure;

(11) FIG. 8 is an exploded view of the exemplary bung locator shown in FIG. 7;

(12) FIG. 9 is a perspective view of the exemplary bung locator of FIG. 7 shown engaging an exemplary drum;

(13) FIG. 10 is a top view of the exemplary bung locator of FIG. 9;

(14) FIG. 11 is a side view of the exemplary bung locator of FIG. 9;

(15) FIG. 12 is a sectional view of the exemplary bung locator of FIG. 11 taken along lines FIG. 12-FIG. 12;

(16) FIG. 13 is an exploded view of the exemplary bung locator of FIG. 12;

(17) FIG. 14 is a top view of an exemplary bung locator engaging an exemplary drum in accordance with one or more embodiments of the present disclosure;

(18) FIG. 15 is a side view of the exemplary bung locator of FIG. 14 before engaging the drum;

(19) FIG. 16 is a side view of the exemplary bung locator of FIG. 14 engaging the drum;

(20) FIG. 17A is a perspective view of the exemplary bung plug extractor shown in FIG. 1A in accordance with one or more embodiments of the present disclosure;

(21) FIG. 17B is a side view of the exemplary bung plug extractor of FIG. 17A in accordance with one or more embodiments of the present disclosure;

(22) FIG. 17C is a cross-sectional view of the exemplary bung plug extractor of FIG. 17B taken along lines FIG. 17C-FIG. 17C;

(23) FIG. 18 is a cross-sectional view of the exemplary bung plug extractor of FIG. 17A in an exemplary home position, or before engaging an exemplary drum, in accordance with one or more embodiments of the present disclosure;

(24) FIG. 19 is an exploded view of an exemplary bung plug engagement tool in accordance with one or more embodiments of the present disclosure;

(25) FIG. 20 is a side view of the exemplary bung plug engagement tool of FIG. 19;

(26) FIG. 21 is an exploded view of another exemplary bung plug engagement tool in accordance with one or more embodiments of the present disclosure;

(27) FIG. 22 is a top view of the exemplary bung plug extractor of FIG. 17A shown engaging an exemplary drum;

(28) FIG. 23 is a perspective view of the exemplary bung plug extractor of FIG. 22;

(29) FIG. 24 is a side view of the exemplary bung plug extractor of FIG. 22;

(30) FIG. 25 is a cross-sectional view of the exemplary bung plug extractor of FIG. 24 taken along lines FIG. 25-FIG. 25;

(31) FIG. 26 is a cross-sectional view of the exemplary bung plug extractor of FIG. 17A contacting an exemplary drum and bung plug therein in accordance with one or more embodiments of the present disclosure;

(32) FIG. 27 is a cross-sectional view of the exemplary bung plug extractor of FIG. 17A contacting an exemplary drum and bung plug at the beginning of extraction of the bung plug in accordance with one or more embodiments of the present disclosure;

(33) FIG. 28 is a cross-sectional view of the exemplary bung plug extractor of FIG. 17A being lifted away from the exemplary drum after removing the exemplary bung plug therefrom in

accordance with one or more embodiments of the present disclosure;

- (34) FIG. 29 is a cross-sectional view of the exemplary bung plug extractor of FIG. 17A in a misaligned position while contacting an exemplary drum and bung plug therein prior to removing the bung plug in accordance with one or more embodiments of the present disclosure;
- (35) FIG. 30 is an isolated side view of certain components of the exemplary bung plug extractor of FIG. 17A including an exemplary bung plug retraction actuator in accordance with one or more embodiments of the present disclosure;
- (36) FIG. 31 is a cross-sectional view of the exemplary bung plug extractor components shown in FIG. 30 taken along lines FIG. 31-FIG. 31;
- (37) FIG. 32 is an exploded view of the exemplary bung plug extractor components shown in FIG. 30;
- (38) FIG. 33 is a perspective view of the vacuum fitting of the exemplary bung plug extractor of FIG. 17A in accordance with one or more embodiments of the present disclosure;
- (39) FIG. 34 is a perspective view of the bung plug conveyor shown in FIG. 1A in accordance with one or more embodiments of the present disclosure;
- (40) FIG. 35 is a perspective view of the exemplary bung plug installer shown in FIG. 1A in accordance with one or more embodiments of the present disclosure;
- (41) FIG. 36 is a cross-sectional view of the exemplary bung plug installer of FIG. 35 in an exemplary home position, or before installing an exemplary bung plug in an exemplary drum, in accordance with one or more embodiments of the present disclosure;
- (42) FIG. 37 is a cross-sectional view of the exemplary bung plug installer of FIG. 35 as it makes contact with an exemplary drum and prior to fully threadably engaging an exemplary bung plug therewith in accordance with one or more embodiments of the present disclosure;
- (43) FIG. 38 is a cross-sectional view of the exemplary bung plug installer of FIG. 35 after threadably engaging an exemplary bung plug too far into the bung opening of an exemplary drum;
- (44) FIG. 39 is a cross-sectional view of the exemplary bung plug installer of FIG. 35 after threadably engaging an exemplary bung plug with an exemplary drum and moving away from the drum in accordance with one or more embodiments of the present disclosure;
- (45) FIG. 40 is an exploded view of various exemplary components of the drum manipulation system of FIG. 1A including the exemplary bung plug installer in accordance with one or more embodiments of the present disclosure;
- (46) FIG. 41 is a perspective view of the exemplary bung plug installer of FIG. 35 shown engaging an exemplary drum;
- (47) FIG. 42 is a top view of the exemplary bung plug installer of FIG. 41;
- (48) FIG. 43 is a side view of the exemplary bung plug installer of FIG. 41;
- (49) FIG. 44 is a cross-sectional view of the exemplary bung plug installer of FIG. 43 taken along lines FIG. 44-FIG. 44;
- (50) FIG. 45 is an isolated side view of certain components of the exemplary bung plug installer of FIG. 35 including the exemplary displacement sensor in accordance with one or more embodiments of the present disclosure;
- (51) FIG. 46 is a top view of the exemplary bung plug installer components of FIG. 45;
- (52) FIG. 47 is an exploded view of the exemplary bung plug installer components of FIG. 45; and
- (53) FIG. 48 is a cross-sectional view of the exemplary bung plug installer components of FIG. 45 taken along lines FIG. 48-FIG. 48.

DETAILED DESCRIPTION OF PRESENTLY PREFERRED EMBODIMENTS

(54) Characteristics and advantages of the present disclosure and additional features and benefits will be readily apparent to those skilled in the art upon consideration of the following detailed description of exemplary embodiments and/or referring to the accompanying figures. It should be understood that the description herein and appended drawings, being of example embodiments, are not intended to limit the claims of this patent or any patent or patent application claiming priority

hereto. On the contrary, the intention is to cover all modifications, equivalents and alternatives falling within the spirit and scope of this disclosure and appended claims. Many changes may be made to the particular embodiments and details disclosed herein without departing from such spirit and scope.

(55) In showing and describing preferred embodiments in the appended figures, common or similar elements are referenced with like or identical reference numerals or are apparent from the figures and/or the description herein. The figures are not necessarily to scale and certain features and certain views of the figures may be shown exaggerated in scale or in schematic in the interest of clarity and conciseness.

(56) As used herein and throughout various portions (and headings) of this patent (including the claims), the terms “invention”, “present invention” and variations thereof are not intended to mean every possible embodiment encompassed by this disclosure or any particular claim(s). Thus, the subject matter of each such reference should not be considered as necessary for, or part of, every embodiment hereof or of any particular claim(s) merely because of such reference.

(57) Certain terms are used herein and in the appended claims to refer to particular components. As one skilled in the art will appreciate, different persons may refer to a component by different names. The use of a particular or known term of art as the name of a component herein is not intended to limit that component to only the known or defined meaning of such term (e.g. bar, rod, cover, panel, spring, plate, bolt). Further, this document does not intend to distinguish between components that differ in name but not function. Also, the terms “including” and “comprising” are used herein and in the appended claims in an open-ended fashion, and thus should be interpreted to mean “including, but not limited to . . .”. Further, reference herein and in the appended claims to components, feature, actions, aspects etc. in a singular tense does not limit the present disclosure or appended claims to only one such component or aspect, but should be interpreted to mean one or more, except and only to the extent as may be expressly limited to one in a particular claim hereof and only for such claim(s) and any claim(s) depending therefrom

(58) As used herein and in the appended claims, the following terms have the following meanings, except and only to the extent as may be expressly specified differently in a particular claim hereof and only for such claim(s) and any claim(s) depending therefrom:

(59) The term “and/or” as used herein provides for three distinct possibilities: one, the other or both. All three possibilities do not need to be available-only any one of the three. For example, if a component is described as “having a collar and/or a coupling”, some embodiments may include a collar, some embodiments may include a coupling and some embodiments may include both. Since the use of “and/or” herein does not require all three possibilities, a claim limitation herein that recites “having a collar and/or a coupling” would be literally infringed by a device including only one or more collars, one or more couplings or both one or more couplings and one or more collars.

(60) The terms “automated” and variations thereof as used herein mean capable of operating or performing one or more tasks without human intervention. One example of automation is with the use of one or more electronic devices.

(61) The terms “bung opening” and variations thereof as used herein mean and include a fill hole or orifice formed in a drum or other container (e.g. in the top of thereof and facing generally upwardly) useful for filling the drum or other container with desired contents and often surrounded by a flange or other closure attachment (e.g. with receiving threads) for receiving a bung plug.

(62) The terms “bung plug” and variations thereof as used herein mean and include one or more caps, covers, closures or plugs releasably engageable with a bung opening, such as by threadable engagement, other mateable engagement, snapping engagement or any other type of releasable engagement and constructed of any suitable material(s) (e.g. steel, plastic, composite, wood, etc.).

(63) The terms “coupled”, “connected” and the like, and variations thereof, as used herein mean either an indirect or direct connection or engagement, except and only to the extent as may be expressly recited and explicitly required in a particular claim hereof and only for such claim(s) and

any claim(s) depending therefrom. Thus, if a first device couples to a second device, that connection may be through a direct connection, or through an indirect connection via other devices and connections (except and only to the extent as may be expressly recited and explicitly required in a particular claim hereof and only for such claim(s) and any claim(s) depending therefrom).

(64) The terms “electronically coupled” and variations thereof as used herein mean in electronic communication, whether by physical connection (e.g. cable, wiring, bus-bar, switch, etc.), non-physical communication (e.g. wireless, such as Wi-Fi, LAN, other), a combination thereof or both, or otherwise.

(65) The terms “deflect” and variations thereof as used herein mean bend, twist, stretch, compress, expand, deform and the like, such as the action of a typical spring.

(66) The terms “drum” and variations thereof as used herein mean and include fifty-five (or more or less) gallon drums, pails, barrels or any other form of container having a bung opening formed therein (e.g. in the top thereof) and constructed of any suitable material(s) (e.g. steel, plastic, woods, etc.).

(67) The terms “drum manipulation operation”, “drum manipulation activity” and variations thereof as used herein mean and include any one or more activities that relate to manipulating a drum such as, without limitation, activities associated with rotating a drum, selectively locating and positioning a bung opening (or bung plug therein) of a drum, removing and/or installing a bung plug in the bung opening of a drum or a combination thereof.

(68) The terms “elongated” and variations thereof as used herein mean, include and refer to an item having an overall length (during the intended use of the item) that is greater than its average width.

(69) The terms “generally”, “substantially” and variations thereof as used herein mean and include (i) completely, or 100%, of the referenced parameter, variable, value, event etc. and (ii) a range of values less than 100% based upon the typical, normal or expected degree of variation or error for the referenced parameter, variable, value, event, etc. in the context of the particular embodiment or use thereof, such as, for example, 90-100%, 95-100% or 98-100%.

(70) The terms “operator” and variations thereof as used herein mean and include one or more humans, robots or robotic components, artificial intelligence-driven components/circuitry, other components and the like.

(71) Any component identified as a “plate” herein includes, but is not limited to, a plate as that term is commonly understood (e.g. a thin, flat sheet or strip of metal or other material, typically used to join or strengthen things or forming part of a machine), and may have non-planar surfaces or construction, may not be thin per se, may have any other form suitable for use in the particular configuration in which it is used (e.g. may be a curved or curvilinear-shaped member, housing, cone, sleeve, flange, collar, etc.) may be comprised of multiple parts or a combination thereof.

(72) The terms “power-driven”, “power-operated” and variations thereof as used herein mean driven or operated by one or more motors, engines, electronic devices or the like to perform one or more subject tasks or activities without manual effort by a human operator to perform the subject tasks or activities.

(73) The terms “rigidly coupled” and variations thereof as used herein mean connected together in a manner that is intended not to allow any, or more than an insubstantial or minimal amount of, relative movement therebetween during typical or expected operations. In other words, if components A and B are rigidly coupled together, they are not movable relative to one another (more than a minimal or insubstantial amount) during typical or expected operations.

(74) As used herein, when a component is “spring-biased”, the component is arranged to be pressed in one general direction by one or more springs and/or other mechanisms, and can be moved back (in the opposite general direction) upon the application of force(s) to the component sufficient to overcome the pressing forces of the spring(s) and/or other mechanism(s). Spring(s) and/or other mechanisms mentioned herein may be referred to as “biasing” the associated component(s) or providing “biasing force(s)” upon or to the associated component(s). The use of the terms “spring-

biased”, “biasing”, “biasing force(s)” and variations thereof herein and in the appended claims does not require the use of one or more actual springs to provide the biasing force(s); any desired or suitable mechanism or arrangement of parts may be used, except and only to the extent as may be expressly recited and explicitly required in a particular claim hereof and only for such claim(s) and any claim(s) depending therefrom.

(75) The terms “spring” and variations thereof as used herein mean and include one or more resilient members (e.g. compression or torsion springs, coil springs, helical springs, Bellville-washers, leaf springs, disc springs) or non-resilient members capable of providing biasing forces upon one or more other members or components. Thus the “spring” may be a spring (in its literal sense) or any other component or combination of components configured to spring-bias one or more other members or components.

(76) The terms “swivel”, “swiveling”, “swiveling movement”, “relative angular movement” and variations thereof as used herein mean moveable around 360° relative to a vertical axis.

(77) The terms “upright”, “vertical”, “vertically-oriented” and variations thereof as used herein mean and include oriented perfectly or substantially vertically, angularly relative to a vertical axis or non-horizontally.

(78) It should be noted that any of the above terms may be further explained, defined, expanded or limited below or in other sections of this patent. Further, the above list of terms is not all inclusive and other terms may be defined or explained below or in other sections of this patent.

(79) Referring initially to FIGS. **1A-2**, an embodiment of a drum manipulation system **10** of the present disclosure is shown. It should be noted at the outset that FIG. **1B** is the same as FIG. **1A** but with some of the subject matter of FIG. **1A** removed to provide clarity to the remaining components. Thus, when reference is made herein to FIG. **1A**, the reader is invited to also refer to FIG. **1B**. The illustrated drum **30** (e.g. FIG. **3**) is a steel drum, but the drum manipulation system **10** can be used with drums constructed of polyethylene, or other forms of thermoplastic or plastic material, or of any other material(s). The illustrated exemplary drum manipulation system **10** includes at least one drum rotator **20** useful to rotate a drum **30** to a desired position, at least one bung locator **40** useful to locate (e.g. and selectively position) at least one bung opening **32** of the drum **30** and/or a bung plug **34** therein (see e.g. FIGS. **3-5**), at least one bung plug extractor **50** useful to remove a bung plug **34** engaged in the bung opening **32** and at least one bung plug installer **60** useful to install a bung plug **34** in the bung opening **32**. However, other embodiments of the drum manipulation system **10** may include any combination of one or more such components and/or any of those components may be combined. For example, some embodiments may not include any one or more of the drum rotator(s) **20**, bung locator(s) **40**, bung plug extractor(s) **50** or bung plug installer(s) **60**. For other examples, some embodiments may include a combined bung locator **40**/bung plug extractor **50**, bung plug extractor **50**/bung plug installer **60**, etc.

(80) Referring still to FIGS. **1A-2**, the exemplary drum manipulation system **10** is shown being used with, or as part of, an exemplary overall drum filling system **70**. The illustrated drum filling system **70** is automated and includes at least one drum filler **74** useful for filling the drums **30** with desired material (e.g. liquid). At least one exemplary drum conveyor **88** is useful to generally carry the drums **30** into and through the drum manipulation system **10** and drum filling system **70**. One or more exemplary dust cap, or cap seal, systems **96** may be used with or incorporated into the drum manipulation system **10** or drum filling system **70**, such as to affix a dust cap over the bung plug **34** after it is installed in the drum **30**. The illustrated drum manipulation system **10** and drum filling system **70** are shown at least partially supported and/or surrounded by a framework **80**. Cables and wiring for various components of the drum manipulation system **10** and drum filling system **70** are shown at least partially contained in cable carriers **72**.

(81) The drum filling system **70**, drum filler **74**, framework **80**, drum conveyor **88** and dust cap system **96** may have any suitable form, construction, components, configuration and operation and are in no way limiting upon the present disclosure or appended claims, except and only to the

extent as may be expressly recited and explicitly required in a particular claim hereof and only for such claim(s) and any claim(s) depending therefrom. For example, the drum filler **74** may include the system disclosed in U.S. Pat. No. 6,543,494 issued on Apr. 8, 2003 to Bellin et al. and entitled "Apparatus & Method for Filling Liquid Containers", the entire contents of which are hereby incorporated by reference herein in its entirety, any version of liquid lance filler (e.g. model DF 5510 single drum filler) offered by the Assignee hereof or any other type of filler. For other examples, the framework **80** may support various components of the drum manipulation system **10** and drum filling system **70** and/or an enclosure **82** having one or more panels **84** (e.g. stainless steel doors with glass) for providing protection, shielding various components of the drum manipulation system **10** and/or drum filling system **70**, allowing selective operator access, any other desired purpose(s) or a combination thereof. For yet a further example, the illustrated drum conveyor **88** may automatically move the drums **30** through the drum manipulation system **10** and drum filling system **70** and may include one or more chain conveyors **90** (e.g. with chains **92a**, **92b**) configured to convey the drums **30** to, between and from one or more positions, or stations, of the drum filling system **70** and/or drum manipulation system **10**. For still another example, the dust cap system **96** may include a vibratory dust cap feeder **97** and cap seal crimping tool **98** or similar components.

(82) Still referring to FIGS. **1A-2**, in the preferred embodiment, the drum manipulation system **10** (and drum filling system **70**) includes three successive linearly-aligned stations **118**, **120**, **122** along the path of the drum conveyor **88**. A first exemplary (e.g. bung plug removal) station **118** is closest to the entrance end **100** of the drum manipulation system **10** and drum filling system **70**, followed by a second (e.g. filling) station **120**, which is followed by a third (e.g. bung plug installation) station **122** closest to the exit end **102** of the drum manipulation system **10** and drum filling system **70**. The exemplary stations **118**, **120**, **122** may be spaced apart as desired. For certain types of drums **30** (e.g. steel 55 gallon drums), for example, the position of a drum **30** at each station may be spaced approximately or exactly 40" apart, but could be spaced closer or farther apart or unevenly spaced apart as desired. Further, some embodiments may include only one or two, or more than three, stations (e.g. four, five, six, etc.) and any desired activities may occur at each station. In the preferred embodiment, one or more sensors **93** (e.g. radial encoders, linear encoders) are associated with the drum conveyor **88** to determine when the drum conveyor **88** has moved the desired distance (e.g. 40") to/from each station **118**, **120**, **122** and assist in causing the drum conveyor **88** to stop, such as via an electric VFD motor **95** or other component(s).

(83) It should be understood that the drum filling system **70** may have more or fewer components than those illustrated and described herein. Additionally, the drum manipulation system **10** of the present disclosure may be used as part of or associated with any other systems, components or apparatus (other than the drum filling system **70**) where it is desirable or necessary to rotate a drum **30**, locate or position a bung opening **32** or bung plug **34** of a drum **30**, remove or replace a bung plug **34** from a drum **30**, or a combination thereof. For example, the drum manipulation system **10** may be used with equipment configured or used for manufacturing, assembling, cleaning, retrofitting, testing, measuring, filling or emptying drums **30**, or may be used alone. Thus the environment, overall system and application with which the drum manipulation system **10** of the present disclosure may be used (including the nature, type, construction, configuration and components of the drum **34**), and the details thereof as provided herein or shown in the appended drawings, are in no way limiting upon the present disclosure or appended claims, except and only to the extent as may be expressly recited and explicitly required in a particular claim hereof and only for such claim(s) and any claim(s) depending therefrom.

(84) Still referring to FIGS. **1A-2**, in some embodiments, the drum manipulation system **10** of the present disclosure may be fully, or nearly-fully, automated to perform the desired drum manipulation operations with minimal or no human intervention. For example, the drum manipulation system **10** may be configured to automatically process a continuing supply of drums

30 with minimal or no human involvement during rotation of each drum **30** to position the bung opening **32** as desired, removal of the bung plug **34** therefrom and installation of a bung plug **34** back in the bung opening **32** (e.g. after filling the drum **30** and/or other activities). If desired, the drum manipulation system **10** may allow other activities to be automatically performed, such as placement and engagement (e.g. crimping via crimping tool **98**) of a dust cap over the bung plug **34**. Absent the occurrence of any fault events, the exemplary drum manipulation system **10** may be configured to run continuously as long as drums **30** are provided on the drum conveyor **88** at the entrance end **100** of the drum manipulation system **10** and removed from the drum conveyor **88** (or directed otherwise) at the exit end **102**. It should be noted that, in various instances, some human involvement may be necessary, such as, for example, to load dust caps into the dust cap system **96**, to perform other tasks or a combination thereof. Further, human involvement may be necessary in response to the occurrence of any, or, certain fault events (e.g. which may stop the operation of the drum manipulation system **10**).

(85) Minimal, limited or no human involvement in the operation of various embodiments of the drum manipulation system **10** during drum manipulation operations may provide one or more benefits, such as reducing manpower requirements and the time and expenses associated therewith, reducing the potential for on-site safety issues, improving the efficiency and effectiveness of drum manipulation operations and processing of drums, etc. Many embodiments of the drum manipulation system **10** could be used to automatically and continuously process a large quantity of drums **30** (e.g. without any or minimal need for on-site personnel), saving extensive time and cost and improving efficiency and capacity, as compared to prior systems and techniques. For example, when the preferred embodiment of the drum manipulation system **10** is used as part of a drum filling system **70** used for filling 55 gallon drums **30** with low viscosity fluid (e.g. water), the drum manipulation system **10** could process approximately one or more drums per minute indefinitely. Other embodiments or applications of the drum manipulation system **10** could process drums **30** at a quicker rate.

(86) Still referring to FIGS. **1A-2**, to provide the desired automation of the drum manipulation system **10**, any suitable configuration, combination and type of components may be used. For example, all, or some, of the components of the drum manipulation system **10** may be electronically controlled (and thus automated) with the use of one or more electronic (e.g. PLC) controllers **12** electronically coupled thereto (e.g. via cable **13**, wireless communication, or in any other manner). In some embodiments, the controller **12** includes at least one personal computer having computer software (e.g. operating system(s), application software, etc.), one or more processors (e.g. microprocessors, CPU), circuitry, screens/displays, etc. In the preferred embodiment, all of the drum manipulation operations of the drum manipulation system **10** may be automatically controlled and/or performed via one or more central controllers **12** without any or minimal human involvement (except as may be specified otherwise herein).

(87) In many applications, particular variables and/or parameters affecting the operation or actions of one or more of the components of the drum manipulation system **10** may need to be set, or introduced, by one or more operators before or during operation of the drum manipulation system **10**. For example, certain data may need to be provided into the controller **12** before the start of drum manipulations operations, such as information relating to the type of bung plug **34**, drum **30** and/or material to be inserted in the drum **30** and the filling of the drum **30**. For another example, in some embodiments, an operator can select (via the controller **12**) the desired torque to be applied to the bung plug **34** during installation based upon the type of bung plug **34**, type of seal on the bung plug **34** (e.g. Teflon, rubber, Viton) or other variable.

(88) In some scenarios, the controller **12** or drum manipulation system **10** may be configured to measure or determine certain variables/parameters or automatically set, or introduce, them without human involvement. In various embodiments, one or more operators may be able to remotely supervise, monitor and control operation of the drum manipulation system **10** and/or adjust one or

more variables/parameters affecting the operation of one or more components of the drum manipulation system **10** through the electronic controller **12** and/or other components. If desired, one or more cameras may be used to assist in off-site monitoring. However, in various embodiments, only some of the drum manipulation operations performed by the drum manipulation system **10** may be automated or the entire system may not be automated.

(89) Still referring to FIGS. **1A-2**, the arrival of the drum **30** at the exemplary first station **118** may be accomplished in any suitable manner. For example, as the drum **30** is transported to the entrance end **100** of the drum manipulation system **10** (and/or drum filling system **70**) on the drum conveyor **88**, one or more sensors **93** may be configured to detect the arrival of the drum **30** at the particular position along the length of the drum conveyor **88** that represents the first station **118** and cause the drum conveyor **88** to stop or otherwise deposit the drum **30** at that location. In the preferred embodiment, upon detection of the drum **30** at the desired position, one or more sensors **93** signal the electronic controller **12**, which signals the drum conveyor **88** to stop moving. If desired, the drum manipulation system **10** may include one or more other sensors **140** (e.g. photo-eye sensors) to confirm (e.g. to the controller **12**) that the drum **30** is at the desired location (e.g. after the drum conveyor **88** stops). This exemplary configuration and operation of parts (e.g. sensor(s) **93**, **140**) may, if desired, be duplicated and/or used similarly for moving the drum **30** to other locations (e.g. the second station **120**, the third station **122**, etc.). However, in some embodiments, the sensor(s) **93** may directly cause the drum conveyor **88** to stop, or any other components may be used to stop the drum conveyor **88** and/or determine when the drum **30** has arrived at a desired location.

(90) Referring still to FIGS. **1A-2**, the drum rotator **20** may have any suitable form, configuration, operation and benefits. For example, the drum rotator **20** may be configured to provide improved control in the rotation of the drum **30** as compared to other known systems. Better control in the rotation of the drum **30** may save time in the processing of the drums **30**, allow for more effective and accurate positioning of the bung opening **32** (e.g. FIG. **3**), automation (without human involvement) of bung opening **32** positioning or bung plug **34** removal or installation operations, provide other advantages or a combination thereof.

(91) In the preferred embodiment, the drum rotator **20** includes front and rear wheeled carriages **110**, **112** typically positioned on the front and rear sides **104**, **106** of the drum manipulation system **10** (on opposing sides of the drum conveyor **88**) and capable of selectively engaging each drum **30** that arrives at the first station **118**. For example, the front wheeled carriage **110** may have one or more “home” positions closest to the front side **104** of the drum manipulation system **10** and the rear wheeled carriage **112** may have one or more “home” positions closest to the rear side **106** of the drum manipulation system **10**. At least one of the exemplary front and rear wheeled carriages **110**, **112** is configured to selectively rotate the drum **30** (e.g. in both directions) to position the bung opening **32** as desired for removal of the bung plug **34** therefrom, filling the drum **30**, installing a bung plug **34**, any other desired purpose(s) or a combination thereof.

(92) The wheeled carriages **110**, **112** may have any suitable form, construction, configuration and operation. Referring particularly to FIG. **6**, for example, each carriage **110**, **112** may include first (outer) and second (inner) upright clamp arms **124**, **126**, each having at least one upper wheel **130** and at least one lower wheel **132**, respectively. While components **130**, **132** are called “wheels” herein and the exemplary embodiments show them as actual wheels in the literal sense, the term “wheel” as used herein encompasses any suitable component(s) useful for engaging and rotating or allowing rotation of the drum **30**, such as a gripper, roller, rotator or similar component. In the preferred embodiment, each clamp arm **124**, **126** is selectively independently movable inwardly and outwardly relative to the drum **30** at the first station **118** between at least first and second positions. In at least one first (home) position, the respective illustrated clamp arm **124**, **126** will be out of the path of any drums **30** on the drum conveyor **88**. In at least one second (engaged) position, one or more of the wheels **130**, **132** of the respective illustrated clamp arm **124**, **126** should contact (engage) the drum **30** at the first station **118**.

(93) The exemplary clamp arms **124, 126** may be movable between positions in any suitable manner. For example, each respective pair of illustrated clamp arms **124, 126** is selectively pivotable within a desired arc around and relative to a pivot pin, or rod, **114** or other component(s) between positions. Preferably, a distinct, selectively controllable driver **44** (e.g. pneumatic or hydraulic cylinder) is configured to be selectively actuated, such as by signal from the controller **12**, to pivot a corresponding clamp arm **124, 126** between positions within a desired range of motion.

(94) However, any other arrangement of components and techniques may be used to rotate the drum **30**. For example, each clamp arm **124, 126** may have only one, or more than two (e.g. three, four, five, etc.), wheels **130, 132** and/or may be moveable in another manner (e.g. linearly) between positions. Moreover, each carriage **110, 112** may include only one, or more than two (e.g. three, four, five, etc.) clamp arms **124, 126** or may not include any clamp arms **124, 126**. Thus, the details of the construction and operation of the carriages **110, 112** and clamp arms **124, 126** are in no way limiting upon the present disclosure or appended claims, except and only to the extent as may be expressly recited and explicitly required in a particular claim hereof and only for such claim(s) and any claim(s) depending therefrom.

(95) Referring again to FIGS. **1A-2**, in some embodiments, before, at or after the arrival of a drum **30** at the first station **118**, the inner clamp arm **126** of at least one of the wheeled carriages **110, 112** may be initially moved inwardly relative to or over the drum conveyor **88** until one or more of the wheels **130, 132** thereon engages the drum **30** at the first station **118**. For example, the inner clamp arm **126** of both wheeled carriages **110, 112** may initially be pivoted inwardly into an engaged position as the drum **30** moves toward them from the entrance end **100** of the drum manipulation system **10** on the drum conveyor **88** and stops upon contacting one or more (e.g. all four) of the wheels **130, 132** thereon. Thus, in such embodiments, the stopping of the drum conveyor **88** to stop movement of the drum **30** at the first station **118** may be generally synchronized, or concurrent, with the drum **30** contacting the wheels **130, 132** of the inner clamp arms **126**. However, in other embodiments, any desired number of wheels **130, 132** of only one inner clamp arm **126** or one or both outer clamp arms **124, 126** may initially engage the drum **30**, or any other combination of wheels **130, 132** of clamp arms **124** and/or **126** or sequence may be used.

(96) In the preferred embodiment, after the wheels **130, 132** of the inner clamp arms **126** engage the drum **30** and the drum **30** stops moving on the drum conveyor **88**, an exemplary rotation platform **94** is raised up to support the drum **30** and lift it off the drum conveyor **88**. When included, the rotation platform **94** may have any suitable form, configuration and operation. For example, the rotation platform **94** may be used to support the drum **30** (off the drum conveyor **88**) while the drum **30** is rotated, include a non-metallic plate (e.g. plastic) to avoid causing sparks during rotation of the drums **30**, avoid causing wear to the drum conveyor **88** (e.g. chains **92a, 92b**) during rotation of the drum **30**, avoid the production of metal shavings due to metal-on-metal friction during rotation of the drum **30**, for any other purpose(s) or a combination thereof. In other embodiments, a different mechanism may be used to lift the drum **30** or the drum **30** may not be lifted.

(97) Still referring to FIGS. **1A-2**, in the preferred embodiment, after the drum **30** is lifted and before it is rotated, the outer clamp arm **124** of at least one of the wheeled carriages **110, 112** can be pivoted inwardly relative to the drum **30** into an engaged position until one or more (e.g. both) of their respective wheels **130, 132** contacts the drum **30**. For example, the outer clamp arms **124** of both carriages **110, 112** may be moved into an engaged position so that all of their wheels **130, 132** contact the drum **30** prior to rotation. Thus, in this embodiment, the wheeled carriages **110, 112** provide a total of up to four points of rotational contact with and around the drum **30** at two different heights on the drum **30** to help secure the desired vertical upright position of the drum, provide effective rotation of the drum (e.g. to help avoid losing a grip on the drum **30** or ability to rotate the drum **30**, avoid drum slippage or shifting), provide improved precision and control of

rotation of the drum **30** (e.g. stopping rotation at a precise position of the drum **30**, slowing rotation in either direction, controlled speed of rotation), for any other purpose(s) or a combination thereof. However, in other embodiments, any combination of the clamp arms **124**, **126** of the wheeled carriages **110**, **112** may be moved into engagement or contact with the drum **30** in any order and at any desired time. For example, both of the clamp arms **124**, **126** of both wheeled carriages **110**, **112** may be moved into engaged positions simultaneously.

(98) If desired, the drum manipulation system **10** may be configured so that one or more clamp arms **124**, **126** of the exemplary front or rear wheeled carriage **110**, **112** are capable of adjusting the position of the drum **30** on the drum conveyor **88**. For example, one or more clamp arms **124**, **126** of the exemplary front and/or rear wheeled carriages **110**, **112** may (e.g. via the associated driver(s) **44**) be selectively actuated to push, or shift, the drum **30** so it is upright, centered over the rotation platform **94** and/or on the drum conveyor **88** (e.g. between chains **92a**, **92b** of the chain conveyor **90**), for any other purpose(s) or a combination thereof.

(99) Still referring to FIGS. **1A-2**, when the drum **30** at the exemplary first station **118** is ready to be rotated, it may be rotated in any suitable manner. For example, at least one of the front and rear wheeled carriages **110**, **112** may be configured to selectively rotate the drum **30**. In the preferred embodiment, at least one of the wheels **130**, **132** on the exemplary front and/or rear wheeled carriages **110** is power-driven and selectively rotated to rotate the drum **30** in a desired direction and for a desired amount of rotation. For example, all four wheels **130**, **132** of the front wheeled carriage **110** may be power-driven (e.g. FIG. **6**), such as by an electronically controllable motor **148** (e.g. electric VFD motor) operatively coupled thereto to selectively rotate all four wheels **130**, **132** and the drum **30** as desired (e.g. in both directions) to provide enhanced rotational effectiveness (e.g. avoid losing grip on the drum **30** or ability to rotate the drum **30**, avoid drum slippage or shifting), improved precision and control of rotation of the drum **30** (e.g. stopping rotation at a precise position of the drum **30**, slowing rotation in either direction, controlled speed of rotation), for any other purpose(s) or a combination thereof. Thus, the exemplary front wheeled carriage **110** represents the powered side of the illustrated drum rotator **20** and the rear wheeled carriage **112** is the non-powered side. The non-powered wheels **130**, **132** of the exemplary drum rotator **20** are free-spinning, such as to assist in or allow rotation of the drum **30**, for any other suitable purpose(s) or a combination thereof. However, in other embodiments, a distinct motor **148**, or other form of driver, may be used to drive one, two or three of the wheels **130**, **132** of the front and/or rear wheeled carriage **110**, **112** (e.g. one driver for each wheel **130**, **132**). Moreover, more or fewer of the wheels **130**, **132** on the front wheeled carriage **110** and/or rear wheeled carriage **112** may be selectively rotated to rotate the drum **30**. Thus, any suitable component(s) may be used to rotate any combination of the wheels **130**, **132** (or other components) of the carriages **110**, **112** in order to rotate the drum **30**, or the drum **30** may be rotated in a different manner (e.g. by rotating the rotation platform **94** or otherwise). Further, the components and technique for rotating the drum **30** are in no way limiting upon the present disclosure or appended claims, except and only to the extent as may be expressly recited and explicitly required in a particular claim hereof and only for such claim(s) and any claim(s) depending therefrom.

(100) Still referring to FIGS. **1A-2**, in the preferred embodiment, the controller **12** directs operation of all of the drivers **44** (to selectively direct movement of the clamp arms **124**, **126**) and the motor **148** (to selectively rotate and stop rotation of the drum **30**). However, in other embodiments, the motor **148** and/or one or more drivers **44** may be controlled by any other components (e.g. sensors) or directly by an operator. If desired, one or both of the exemplary carriages **110**, **112** or one or more components thereof may be selectively adjustable (e.g. by the controller **12**, manually, etc.) to accommodate different sizes and types of drums **30**, for any other purpose(s) or a combination thereof. For example, the positions, or range of motion, of any of the clamp arms **124**, **126** may be adjustable, the spacing between the clamp arms **124**, **126** may be adjustable, the height of the wheels **130**, **132**, and/or spacing between wheels **130**, **132** may be adjustable or a combination

thereof. Furthermore, one or more components (e.g. the motor **148**) of the drum rotator **20** may be selectively controllable (e.g. via the controller **12** and/or one or more operators) to control the timing, direction, speed and amount of rotation of the drum **30**, or a combination thereof.

(101) The exemplary drum rotator **20** provides one or more advantages. In some embodiments of the drum rotator **20**, the use of a total of eight wheels (e.g. four upper wheels **130** and four lower wheels **132**), allows for more points of contact with the drum **30** and more effective and reliable drum rotation as compared to other drum rotation systems having fewer wheels. For example, when one or more wheels encounters an uneven outer surface (dents, concave portions) of the drum **30**, the wheel could become ineffective at rotating the drum **30** and the drum **30** could stop or slip. In some embodiments of the drum rotator **20**, the use of four independently moving clamp arms **124**, **126** provides for improved engagement with and positioning of the drum **30** because each clamp arm **124**, **126** can independently engage the drum **30**. One or more of the clamp arms **124**, **126** can be used to move or push the drum **30** into proper engagement with other clamp arms **124**, **126** and/or center, or position, the drum **30** in the precise desired location (e.g. within 1/32" or more or less). For another example, the use of all (e.g. four) wheels **130**, **132** of the front wheeled carriage **110** as power-driven wheels may provide for more reliable rotation of the drum **30** (e.g. less chance for slippage) and better speed control of drum rotation (e.g. to ramp up and/or down the speed of rotation).

(102) Referring again to FIGS. **1A-2**, as the drum **30** is rotated, the exemplary bung locator **40** is configured to locate at least one bung opening **32** formed in an upper surface, or top, **36** (e.g. FIG. **3**) of the drum **30** (and/or a bung plug **34** therein), position the bung opening **32** (and/or bung plug **34** therein) as desired for further operations or both. For example, the bung locator **40** may be selectively positioned at least partially over the top **36** of the drum **30** as the drum **30** is rotated (e.g. FIGS. **9-16**). In various embodiments, the bung locator **40** may be initially positioned at a desired location over the top **36** of the drum **30** or shuttled or moved into (and out of) a desired position, or the drum **30** may be moved relative to the bung locator **40**. When the bung locator **40** is moved relative to the drum **30**, any suitable components and technique may be used. For example, one or more drivers **46** may be operatively coupled to the bung locator **40** (e.g. via one or more bracket mounts **42**, FIG. **7**) for selectively shuttling the bung locator **40** in one or more linear, arc-like, angular or other paths back and forth between at least one "home" position (away from the drum **30**) and at least one "operating" position (proximate to the drum **30**). In a preferred "home" position, the exemplary bung locator **40** is sufficiently clear of the drum **30** to allow one or more other components (e.g. bung plug extractor **50**) to be shuttled into position over the drum **30**, to allow the drum **30** to be moved to a different location, for any other desired purpose(s) or a combination thereof. In a preferred "operating" position, the exemplary bung locator **40** is positioned so that when the bung opening **32** (and/or bung plug **34** therein) of the rotating drum **30** is at a particular position below it (e.g. FIGS. **14-16**), the drum **30** will be in the desired position for further operations, such as bung plug **34** removal, bung plug **34** installation, drum filling, testing, other operation(s) or purpose(s) or a combination thereof. In the preferred embodiment, the bung locator **40** is moved into the preferred operating position during rotation of the drum **30**, but could instead be moved before rotation of the drum **30**.

(103) Still referring to FIGS. **1A-2**, in the preferred embodiment, the bung locator **40** is moveable between home and operating positions by a first driver **46a** (e.g. linear pneumatic or hydraulic cylinder) in a linear path sideways (e.g. horizontally) relative to the drum **30** and a second driver **46b** (e.g. linear pneumatic or hydraulic cylinder, FIG. **1C**) in a linear path up and down relative to the drum **30**. The exemplary drivers **46** are preferably electronically coupled to and controlled by the electronic controller **12**. For example, the bung locator **40** may be moved from one or more preferred "home" positions first in a horizontal path toward the drum **30** by actuation of the first driver **46a** and then in a vertical path down to one or more preferred "operating" positions by actuation of the second driver **46b**. It should be noted that the opposite sequence could instead be

used. Further, the same technique and components may be used when it is desired to move the exemplary bung locator **40** away from the drum **30** but in the opposite order or direction (e.g. after locating and positioning the bung opening **32** or at any other desired time). Also, a similar arrangement of parts and techniques can be used to shuttle other components (e.g. the exemplary bung plug extractor **50** and bung plug installer **60**) toward and away from the drum **30** as desired. However, in other embodiments, the bung locator **40** (bung plug extractor **50** and bung plug installer **60**) may be moveable toward and away from the drum **30** in any other desired manner, along only a single path or one or more non-linear paths (e.g. an arc-like motion or angular motion), with the use of any other form or combination of driver(s) or other components controlled by the controller **12**, other component(s) (e.g. sensor(s)), one or more operators or a combination thereof. Thus, the details, nature and characteristics of components used for shuttling, or moving, the bung locator **40** (bung plug extractor **50** and bung plug installer **60**) are not limiting upon the present disclosure and appended claims, except and only to the extent as may be expressly recited and explicitly required in a particular claim hereof and only for such claim(s) and any claims depending therefrom.

(104) The bung locator **40**, such as shown in FIGS. 7-16, may have any suitable form, configuration, components and operation and may locate the bung opening **32** (and/or bung plug **34**) in any suitable manner. Referring particularly to FIGS. 7, 8 & 13, preferably, the bung locator **40** includes a carrier **150** that houses at least one bung location sensor **154** configured to sense the location of the bung opening **32** (or bung plug **34** therein) beneath it. For example, referring to FIG. 13, the carrier **150** may carry first and second downwardly facing, fiber optics sensors **154a**, **154b**, which are spaced-apart the same distance as the distance between opposing edges **35a**, **35b** of an upwardly protruding (e.g. outer) lip **35** of the bung opening **32** or, in some instances, opposing edges **37** (e.g. FIG. 4) of the bung plug **34**. However, any number (e.g. 1, 3, 4, etc.) of any type of bung location sensors **154** may be used in any desired configuration and with or without the use of a carrier **150**.

(105) Referring to FIGS. 7 and 8, the carrier **150**, when included, may have any suitable form, configuration and operation. In the preferred embodiment, the carrier **150** includes a main body **166** having a respective orifice, or slot, **167** formed therein for each bung location sensor **154** and that allows light beams to pass between the corresponding bung location sensor **154** and the top **36** of the drum **30** (e.g. FIG. 13). For example, the illustrated main body **166** is a generally rigid (e.g. metal), horizontally-extending plate, but could have any other form and configuration. If desired, the carrier **150** may include one or more stabilizing members **152** to add rigidity to the main body **166**, help maintain the desired position of the bung location sensor(s) **154**, for any other suitable purpose(s) or a combination thereof. For example, the illustrated embodiment shows a first stabilizing member **152a** in the form of a generally rigid (e.g. metal) plate extending upwardly from the top and across substantially the entire length of the main body **166**, and a second stabilizing member **152b** in the form of a generally rigid (e.g. metal) plate extending upwardly between the top of the main body **166** and a mounting plate **153** coupled to or integral with the main body **166**. However, the stabilizing member(s) **152**, when included, may have any other suitable form, configuration and location.

(106) In some embodiments, one or more sensor guards **156** may be provided proximate to the bung location sensors **154** to assist in shielding them from damage due to inadvertent contact with other items, for any other purpose(s) or a combination thereof. The sensor guards **156** may have any suitable form, configuration and operation. For example, the sensor guard(s) **156** may protect the bung location sensors **154** from contact with tools or other items left on the top **36** of the drum **30**, which could damage or destroy the sensors **154** as the drum **30** is rotated under the bung locator **40**. In the preferred embodiment, the bung locator **40** includes a single sensor guard **156** affixed to the bottom of the main body **166**, such as with releasable connectors **157** (e.g. screws, bolts, pins, etc.), and extending at least partially around the bung location sensors **154**. The exemplary sensor

guard **156** is shown having a slot **155** into, or through, which the bung location sensors **154** extend or project light. The sensor guards **154** may also be constructed of any suitable material, such as durable and/or lightweight material (e.g. plastic, aluminum). In other embodiments, the sensor guard(s) **156**, when included, may be integral with the carrier **150** (or other component), include multiple parts or have any other form, configuration, operation and function.

(107) Any suitable “operating position” of the bung locator **40** may be used. In various embodiments, the bung locator **40** is configured to ride on the outer rim, or chime, **31** (or other part) of the drum **30** (e.g. FIGS. **9-12**) to assist in accurately positioning the carrier **150** and/or bung location sensor(s) **154** carried thereby in one or more operating positions over the drum **30** as desired and/or for any other purposes. For example, as shown in FIGS. **7 & 8**, the carrier **150** may include one or more feet **162** configured to rest (and ride) upon the outer rim **31** of the drum **30** after the bung locator **40** is lowered into the desired position over the drum **30**. In the preferred embodiment, the main body **166** of the carrier **150** includes at least first and second spaced-apart feet **162a**, **162b** (e.g. plastic shoes) extending downwardly from the main body **166**. For example, the feet **162a**, **162b** may be (i) releasably rigidly coupled (e.g. with bolts, pins or the like) to the main body **166** (e.g. at or proximate to the respective ends **166a**, **166b** thereof) or integral therewith, (ii) configured to rest and ride upon the outer rim **31** (e.g. FIGS. **9-11**, **14**), or another part, of the drum **30**, (iii) spaced a desired distance D_i (e.g. FIG. **15**) radially outwardly relative to the bung location sensors **154**, (iv) formed with a desired height H_1 (e.g. FIG. **15**) to accurately position the bung location sensors **154** over the bung opening **32** (and/or bung plug **34**) of the rotating drum **30**, or a combination thereof.

(108) Still referring to FIGS. **7 & 8**, if desired, the carrier **150** may be configured to effectively float, or be spring-biased downwardly, on the outer rim **31** (or other part) of the drum **30** to allow it to engage drums **30** having (or at) different heights, accommodate misalignment of the bung locator **40**, accommodate misalignment or imperfections of the drum **30** or misalignment of its position as the drum rotates, avoid jamming or damaging the bung locator **40** or disrupting operations, for any other purpose(s) or a combination thereof. This may be accomplished in any suitable manner. In the preferred embodiment, the carrier **150** effectively floats via one or more linear bearings **170** rigidly coupled thereto and movable on an upright linear bearing rail **172** (e.g. mounted to bracket mount **42**). For examples, FIG. **16** shows the exemplary carrier **150** of FIG. **15** after being lowered down into an operating position (with its feet **162** riding on the outer rim **31** of the drum **30**) and floating up on the linear bearing rail **172** via the linear bearing **170**. However, any other suitable components (e.g. springs, flexible members, shock absorbers, etc.) and techniques may be used to float or spring-bias the carrier **150** downwardly relative to the drum **30** and/or provide one or more operating positions of the bung locator **40**. Further, the bung locator **40** may have any other suitable form configuration, components and operation. For example, the bung locator **40** may, in some embodiments, ride on the top **36** of the drum **30** (instead of the outer rim **31**).

(109) Referring now to FIGS. **7 & 13**, the bung location sensors **154** may have any suitable form, configuration and operation. In the preferred embodiment, light beams are projected down through the exemplary fiber optics sensors **154a**, **154b** to the top **36** of the drum **30** and reflected up from the drum **30** through the sensors **154a**, **154b** to determine the presence of the bung opening **32** (or bung plug **34**) at the desired location below the bung locator **40**. This may be performed in any suitable manner. For example, one or more optic amplifiers **158** (and/or or other suitable components) may be associated with or coupled to the exemplary sensors **154a**, **154b** (e.g. with wires **160**) to (i) cause one or more (e.g. red) light signals to project down through each sensor **154a**, **154b** onto the top **36** of the rotating drum **30**, (ii) receive light signal(s) reflected up off the top **36** the drum **30** through the sensors **154a**, **154b**, (iii) determine when reflected light signals received through both sensors **154a**, **154b** indicate the sensors **154a**, **154b** are simultaneously positioned over respective opposing edges **35a**, **35b** of the lip **35** of the bung opening **32** (or similar

edges of the bung plug **34**) (e.g. based upon amplitude and/or intensity of the reflected light signals), (iv) signal the electronic controller **12** (other component or operator) to cease rotation of the drum **30**, for any other purpose(s) or a combination thereof. However, other embodiments may include only one or more than two (e.g. three, four, etc.) bung location sensors **154** or other components for determining the position of the bung opening **32** (and/or bung plug **34**). Further, any other types, or combinations of types, of bung location sensors **154** may be used (e.g. electronic proximity sensors, inductive sensors, image sensors (e.g. cameras)). Moreover, the bung location sensors **154** may communicate with the electronic controller **12**, motor **148** (e.g. FIG. 1A) other component or one or more operators (e.g. without the use of an optic amplifier **158**) directly or through one or more other component(s)).

(110) In accordance with various embodiments of the drum manipulation system **10**, the bung locator **40** may provide one or more advantages. For example, by engaging the chime **31** instead of the top **36** of the drum **30**, various embodiments of the bung locator **40** will avoid scratching, creating wear marks or other visual impairments to the top **36** of the drum **30**. In contrast, the chime **31** of many versions of drums **30** is intended to be scratched, nicked, etc. For another example, the bung locator **40** may be lightweight and engage the drum **30** with minimal friction, improving the longevity of and reducing the occurrence of wear to the bung locator **40** (e.g. feet **162**) and the part of the drum **30** (e.g. chime **31**) engaged thereby. In the preferred embodiment, the main body **166** and stabilizing members **152** of the carrier **150** are constructed of lightweight aluminum plates, the bung location sensors **154** are lightweight and the sensor guards **156** are lightweight plastic. For yet another example, the bung locator **40** may be easily assembled and repaired, constructed with few or no moving or mechanical parts, have minimal parts that could malfunction, wear or become damaged or a combination thereof. For still a further example, the embodiments of the bung locator **40** not having components (e.g. wheels) that ride across the top **36** of the drum **30**, will not pick up dirt, debris, liquid and like from the top **36** of the drum **30** and become damaged or clogged thereby and/or malfunction.

(111) Referring back to FIGS. 1A-2, in the preferred embodiment, after the desired rotation of the drum **30** and positioning of bung opening **32** and/or bung plug **34**, the exemplary bung locator **40** may be selectively shuttled, or moved, away from the drum **30**. For example, the bung locator **40** may be moved back to one or more home positions, such as in the same or similar manner described above (e.g. in the reverse order). If the drum **30** was previously raised up off the drum conveyor **88**, it may be lowered back down onto the drum conveyor **88**. In the preferred embodiment, the exemplary rotation platform **94** is lowered to set the drum **30** back down on the drum conveyor **88**. However, there may be instances or embodiments in which the drum **30** remains in a raised position, is not lowered down onto the drum conveyor **88**, is moved to a different location or a combination thereof.

(112) The exemplary bung plug extractor **50** (e.g. FIGS. 17A-33) will now be described. Referring to FIGS. 1A-2, to remove a bung plug **34** from the drum **30** with the drum manipulation system **10**, such as after the bung opening **32** has been suitably positioned, the exemplary bung plug extractor **50** may be shuttled, or moved, into position relative to the drum **30**, the drum **30** may be moved in a desired position relative to the bung plug extractor **50**, any other technique may be used to position the bung plug extractor **50** or a combination thereof. For example, the bung plug extractor **50** may be moved from at least one “home” position (e.g. away from the drum **30**, similar to the home position of the exemplary bung locator **40** as described above) to an “operating” position over and/or in contact with the drum **30** (e.g. FIGS. 22-25). In such instances, the bung plug extractor **50** may be selectively moveable relative to the drum **30** in any suitable manner. For example, one or more drivers **46** may be operatively coupled to the bung plug extractor **50** and, if desired, electronically coupled to electronic controller **12**, for selectively moving the bung plug extractor **50** in one or more linear, arc-like and/or angular paths back and forth between positions, similarly as described above with respect to the movement of the exemplary bung locator **40**. In the

preferred embodiment, the bung plug extractor **50** is moveable between home and operating positions by the first driver **46a** (e.g. linear pneumatic or hydraulic cylinder; FIG. **1C**) in a first linear path sideways (e.g. horizontally) toward the drum **30** and then by the second driver **46b** (e.g. linear pneumatic or hydraulic cylinder) in a linear vertical path down to the drum **30**. However, in other embodiments, the bung plug extractor **50** may be moveable toward and away from the drum **30** in any other desired manner, along a single path or one or more non-linear paths (e.g. an arc-like motion, angular motion, etc.), with the use of any other form or combination of driver(s) or other components controlled by the controller **12**, other component(s) (e.g. sensor(s)) or one or more operators, or a combination thereof.

(113) The bung plug extractor **50** may have any suitable form, configuration, components and operation. Referring to FIGS. **17A-18**, in the preferred embodiment, for example, the bung plug extractor **50** includes a bung plug engagement tool **184**, a centering assembly **180** and a frame, or carrier, **200**. The exemplary bung plug engagement tool **184** is useful to disconnect the bung plug **34** from the drum **30**, the illustrated centering assembly **180** is useful to assist in centering the bung plug engagement tool **184** over the bung opening **32** and the carrier **200** carries the centering assembly **180** and bung plug engagement tool **184**. However, these components may have one or more additional or different purposes, or a combination thereof. Further, one or more of these components may be combined and some embodiments may include only one or two of these components or additional components.

(114) Referring still to FIGS. **17A-18**, the carrier **200** may have any suitable form, configuration, components and operation for carrying the centering assembly **180** and bung plug engagement tool **184**. In many embodiments, the carrier **200** includes at least one outer spring support rod, or linear bearing shaft, **238** engaged between at least one upper plate **239** and at least one lower plate **260**. In such embodiments, at least one outer spring **236** (e.g. coil spring, helical compression spring) extends around each respective rod **238** and biases at least one exemplary floating plate **254** downwardly on the spring support rod(s) **238** toward and relative to the lower plate **260**. However, the components of the carrier **200** may vary in form from such embodiments. For example, any of the plates **239**, **254**, **260** may not be plates per se, but instead be in any other desired form and may be comprised of multiple parts. Moreover, the carrier **200** may include only some of these components and/or may have additional or different components and functions. Thus, the form, components, configuration and operation of the carrier **200** are in no way limiting upon the present disclosure or appended claims, except and only to the extent as may be expressly recited and explicitly required in a particular claim hereof and only for such claim(s) and any claim(s) depending therefrom.

(115) When included, the bung plug engagement tool **184** may have any suitable form, configuration, components and operation for assisting in disconnecting the bung plug **34** from the drum **30**. For example, referring to FIGS. **18-20**, the bung plug engagement tool **184** may include at least one bung plug mating portion, or wrench, **192** disposed at its lower end and configured to engage the bung plug **34** (e.g. FIG. **26**) and disconnect it from the drum **30**, and at least one guide **196** extending upwardly from the mating portion **192**. The mating portion(s) **192** may have any suitable form, configuration, components and operation. In instances in which the bung plug **34** is threadably engaged with the drum **30** (e.g. rotated clockwise for engagement and counterclockwise for removal), the mating portion **192** may be used to rotate the bung plug **34** in any suitable manner. For example, the mating portion **192** may include at least one tab **194** extending (e.g. downwardly) therefrom and configured to engage and facilitate rotation of the bung plug **34**.

(116) The tab(s) **194** may have any suitable form, configuration and operation. In the preferred embodiment, the mating portion **192** includes first and second tabs **194** each engageable with a distinct protrusion **38** (e.g. FIG. **26**) extending (e.g. upwardly) from the bung plug **34**. For example, the first and second tabs **194** may be configured to be moved into abutting contact with the respective protrusions **38** of the bung plug **34** (e.g. metallic bung plug **34a**, FIG. **4**) to push them in

a rotational path to rotate the bung plug **34**. In other embodiments, the mating portion **192** may include only one, or more than two, tabs **194** as necessary (e.g. depending upon the configuration of the bung plug **34**). For example, the bung plug engagement tool **184** may have four tabs **194** that engage four respective protrusions **38** of bung plug **34** (e.g. plastic bung plug **34b**, FIG. 5). If desired, at least one (e.g. inner) edge **194a** of each tab **194** may be tapered, such as to properly register with, or engage, the bung plug **34** (e.g. plastic bung plug **34b**) and/or for any other desired purpose(s).

(117) In other embodiments, the tab(s) **194** (or other components) of the mating portion **192** may instead physically or mechanically mate with, grip or otherwise engage the bung plug **34** to facilitate engagement with the bung plug **34** (e.g. for disconnection from or installation in the drum **30**). For example, if the bung plug **34** includes one or more socket-like recesses having a particular cross-sectional shape (e.g. square, hexagonal, octagonal, rectangular, slotted, etc.), the mating portion **192** of the bung plug engagement tool **184** may include one or more tabs **194** in the form of a bit, pin, blade, spade, etc. (e.g. having a complimentary cross-sectional shape, such as square, hexagonal, rectangular, octagonal, etc.) that are insertable into the socket-like recess(es) of the bung plug **34**. For another example, if the bung plug **34** includes one or more protrusions **38** (e.g. bolt heads, bits, pins, blades, spades, etc.) having a particular cross-sectional shape (e.g. square, hexagonal, octagonal, rectangular, etc.), the mating portion **192** of the bung plug engagement tool **184** may include one or more socket-like recesses having a complimentary cross-sectional shape and being engageable around the corresponding protrusion(s) **38**. For still further examples, the mating portion **192** may instead or also include one or more clips, claws, grippers, suction members or the like for engaging the bung plug **34** (e.g. for disconnection from or installation in the drum **30**).

(118) Still referring to FIGS. **18-20**, the guide(s) **196**, when included as part of the bung plug engagement tool **184**, may have any suitable form, configuration, components and operation. For example, the guide **196** may be an elongated control rod **198** coupled to or integral with the mating portion **192** and configured to assist in moving the mating portion **192** as desired to disconnect the bung plug **34**. In the preferred embodiment, the guide **196** includes a mounting plate **202** (e.g. coupled thereto by weld and/or shrink-fitting) to which the mating portion **192** is mounted.

(119) In embodiments configured to disconnect a rotatable bung plug **34** from the drum **30**, the exemplary guide **196** may be secured to, or integral with, the mating portion **192** and selectively rotatable to rotate the mating portion **192**. For example, the exemplary guide **196** may be operatively coupled to a driver **204** (FIG. **17A**) for selective concurrent rotation of the guide **196** and mating portion **192** in either direction (e.g. counterclockwise for unscrewing the bung plug **34** and clockwise for installing the bung plug **34**). As the preferred bung plug engagement tool **184** is rotated during normal operating conditions, sufficient rotational pressure and/or torque should be applied by the mating portion **192** to the bung plug **34** (e.g. FIG. **27**) to unscrew it from the drum **30**. If the exemplary guide **196** and mating portion **192** are not integrally formed, they should preferably be sufficiently rigidly coupled together to ensure they withstand the expected forces applied thereto during rotation of the bung plug **34**. For example, the mating portion **192** may be releasably rigidly coupled to the guide **196** (e.g. via mounting plate **202**) with one or more (e.g. two) pins **199** (or other connectors) to provide the desired rigid connection, for any other purpose(s) or a combination thereof. However, the mating portion **192** and guide **196** may include any other arrangement of components and configuration. For example, if the mating portion **192** is configured to grip the bung plug **34**, the guide **196** may include an actuator (e.g. handle, trigger, up-and-down sliding member(s), etc.) to move the mating portion **192** into and out of gripping engagement with the bung plug **34**.

(120) Referring to FIGS. **19 & 20**, in some embodiments, the bung plug engagement tool **184** (or one or more portions thereof) may be removable from the drum manipulation system **10**, such as to switch out match the mating portion **192** with the particular type of bung plug **34** being removed or

drum **30** being serviced, for maintenance, repair or replacement, for any other purpose(s) or a combination thereof. For example, the bung plug mating portion **192** may comprise a bung plug engagement plate **195** that is releasable from the bung plug engagement tool **184** via the pins **199** (e.g. dowel pins) or other releasable connectors. In some embodiments, it may be required or desirable to use a mating portion **192** (e.g. engagement plate **195**) that is constructed at least partially of non-spark material (e.g. bronze), such as when filling the drum with flammable fluid or materials. Thus, a typical metallic mating portion **192** may be switched out and replaced with a non-spark version as desired. However, the bung plug engagement tool **184** may have any other configuration of components and operation. Thus, the details, nature and characteristics of the bung plug engagement tool **184** are not limiting upon the present disclosure and appended claims, except and only to the extent as may be expressly recited and explicitly required in a particular claim hereof and only for such claim(s) and any claims depending therefrom.

(121) Referring now to FIGS. **17A-18**, when included, the driver **204** for actuating the exemplary guide **196** of the bung plug engagement tool **184** may have any suitable form, configuration and operation. In many embodiments, the driver **204** is an air motor **206** having a rotatable shaft **208** operatively coupled to the upper end of the guide **196** and configured to rotate it. However, any other forms of the driver(s) **204** (e.g. pneumatic or hydraulic cylinder, servo-motor, etc.) may be used for actuating the guide **196**. The exemplary driver **204** is controlled by the electronic controller **12**, but may be controlled by any other component(s), one or more operators or a combination thereof.

(122) If desired, one or more flexible couplers **212** may be coupled between the driver **204** (e.g. air motor **206**) and the guide **196** to help facilitate selective movement of the guide **196**, provide flexibility in engaging the bung plug engagement tool **184** with the bung plug **34**, help avoid jamming or damaging the bung plug **34**, the bung plug engagement tool **184** or other components and/or disrupting operations during engagement and positioning of the bung plug engagement tool **184**, for any other purpose(s) or a combination thereof. When included, the flexible coupler(s) **212** may have any suitable form, configuration and operation. For example, the flexible coupler **212** may include one or more universal joints **214** disposed between the driver **204** and the guide **196**. In the preferred embodiment, the flexible coupler **212** includes a universal joint **214** having an upper link **215** for engaging the shaft **208** of the driver **204** and a lower link **216** for engaging the guide **196** (e.g. control rod **198**). The illustrated universal joint **214** may be configured, for example, to allow the guide **196** the ability to “flex”, or pivot, around 360° up to approximately ¼” displacement relative to the vertical axis **220** of the bung plug extractor **50** and/or the drum **30**. If desired, the upper end **196a** (e.g. FIG. **17C**) of the guide **196** may be slideable up and down within a bore **213** of the flexible coupler **212** in a limited range of motion to allow some limited up-and-down movement of the guide **196** relative thereto and to the drum **30**. However, other embodiments may involve the use of one or more flexible couplers **212** (e.g. universal joints **214**) or other components that allow the guide **196** to move in a range of motion less than 360° and/or more or less than ¼” displacement (e.g. ⅛”, ½”, ¾”, 1”, etc.), or do not allow flexing or up-and-down movement of the guide **196** relative to the coupler **212**. Moreover, the flexible coupler(s) **212**, when included, may have any other suitable form and operation. Thus, the details, nature and characteristics of the flexible coupler(s) **212** are not limiting upon the present disclosure and appended claims, except and only to the extent as may be expressly recited and explicitly required in a particular claim hereof and only for such claim(s) and any claims depending therefrom.

(123) Referring still to FIGS. **17A-18**, the centering assembly **180**, when included in the bung plug extractor **50**, may have any suitable form, configuration, components and operation to assist in centering the bung plug engagement tool **184** over the bung opening **32** (e.g. FIG. **26**). For example, the centering assembly **180** may include a centering collar **228** configured to at least partially surround the mating portion **192** of the bung plug engagement tool **184** and assist in positioning the bung plug engagement tool **184** over the bung plug **34** as desired. In the preferred

embodiment, the mating portion **192** is essentially nested within the centering assembly **180** and the centering collar **228** substantially or entirely surrounds the mating portion **192** and is configured to rest at least partially upon, or engage, the lip **35** of the bung opening **32** (e.g. FIG. **26**). When the exemplary centering collar **228** engages the lip **35** around the circumference of the bung opening **32** during normal operating conditions, the bung plug engagement tool **184** should typically be properly (e.g. perfectly) aligned over the bung plug **34** for disconnecting it from the drum **30**.

(124) Referring now to FIGS. **18** & **26**, if desired, the centering collar **228** may be configured to sit substantially perfectly flat on the lip **35** of the bung opening **32** (or other part of the drum), form a partial, substantial or entire fluid-tight seal around the bung opening **32** (e.g. with the lip **35**), for any other purpose(s) or a combination thereof. In some embodiments, the centering collar **228** may include a taper **229** around its inner edge **233** and/or a ledge **231** formed inwards of the inner edge **233** to match, mate with or seat around the circumference of the lip **35**, such as to help the centering collar **228** sit substantially perfectly flat on the lip **35**, form a substantially or entirely fluid-tight seal with the lip **35** around the bung opening **32**, for any other purpose(s) or a combination thereof. In other embodiments, the centering collar **228** may engage any other part (e.g. the top **36**) of the drum **30** to achieve the same or other purpose(s). Moreover, the centering collar **228** may have any other configuration and operation. Thus, the details of the construction and operation of the centering collar **228** are in no way limiting upon the present disclosure or appended claims, except and only to the extent as may be expressly recited and explicitly required in a particular claim hereof and only for such claim(s) and any claim(s) depending therefrom.

(125) Referring to FIGS. **26** & **27**, once the preferred bung plug engagement tool **184** and centering collar **228** are moved into engagement with the bung plug **34** and drum **30** (e.g. FIG. **26**), respectively, the bung plug engagement tool **184** and/or centering collar **228** may be maintained in such engagement in any suitable manner. For example, the centering collar **228** and/or bung plug engagement tool **184** may be spring-biased to assist in maintaining the desired engagement. In such instances, any suitable components may be used to spring-bias the bung plug engagement tool **184** and/or centering collar **228** as desired. In the exemplary embodiment, one or more (e.g. three) of the outer springs **236** provide downward spring-biasing forces on the centering collar **228** to assist in maintaining its engagement (e.g. contact) with the drum **30** (e.g. lip **35**) during bung plug **34** removal, and one or more (e.g. three) inner springs **240** (e.g. coil spring) provide downward spring-biasing forces on the bung plug engagement tool **184** to assist in maintaining its engagement (e.g. contact) with the bung plug **34** during bung plug removal. However, in other embodiments, any other quantity and arrangement of one or more springs, or one or more than two sets of springs, or other biasing members, may be used. Moreover, some embodiments may not require any springs, biasing members and/or may include other or additional components to maintain the desired engaged relationship of the bung plug engagement tool **184** with the bung plug **34** and/or the centering collar **228** with the drum **30**. For example, in some embodiments, one or more drivers (e.g. motors) may be used to retain or hold the bung plug engagement tool **184** and/or centering collar **228** in the desired engaged position(s).

(126) Referring now to FIGS. **18**, **26** & **27**, in various embodiments, it may be desirable to provide limited positional flexibility (e.g. angular, up-and-down, etc.) of the centering collar **228** and/or bung plug engagement tool **184** relative to the drum **30** while seating the centering collar **228** on the drum **30** and/or bung plug engagement tool **184** on the bung plug **34**. This may be desirable, for example, to allow precision in properly aligning and seating the centering collar **228** and/or bung plug engagement tool **184** to (i) improve performance of the drum manipulation system **10** (e.g. bung plug removal and installation) and/or efficiency and effectiveness of operations, (ii) assist in preventing the centering collar **228** and/or the bung plug engagement tool **184** from undesirably hitting or contacting the drum **30** or bung plug **34** and any resulting binding, jamming or damaging of components or disruption of operations, (iii) allow proper seating of the centering collar **228**

and/or bung plug engagement tool **184** in imperfect situations (e.g. the drum **30** is not perfectly upright, the top **36** of the drum **30** is not perfectly horizontally-oriented or flat, the drum **30** or bung plug extractor **50** is misaligned, the drum **30** inadvertently slid or shifted position during lowering onto the drum conveyor **88**), for any other purpose(s) or a combination thereof. Any suitable components may be used to allow positional flexibility of the centering collar **228** and/or bung plug engagement tool **184** relative to the drum **30**. For example, the centering collar **228** and/or the bung plug engagement tool **184** may be provided with upward (or up-and-down) positional flexibility relative to the drum **30**. In the preferred embodiment, the (e.g. three or more or less) outer spring(s) **236** acting upon the centering collar **228** allow limited up-and-down movement of the centering collar **228** relative to the drum **30**, and the exemplary (e.g. three or more or less) inner spring(s) **240** acting upon the bung plug engagement tool **184** allow limited up-and-down movement thereof relative to the drum **30** (and centering collar **228**). However, any other configuration of components (elastic members, shock absorbers, bearings, etc.) may be used to provide upward positional flexibility of the centering collar **228** and/or bung plug engagement tool **184**, if this feature is included.

(127) Still referring to FIGS. **18**, **26** & **27**, in some embodiments, the drum manipulation system **10** may be configured to limit the up and/or down movement of the centering collar **228** and/or the bung plug engagement tool **184** relative to other components. For example, while the upper end **196a** of the exemplary guide **196** of the bung plug engagement tool **184** is moveable up and down within the bore **213** of the flexible coupler **212**, the guide **196** may include a ledge **197** (or other feature) that shoulders-up against, or contacts, the lower end **217** (or other part) of the flexible coupler **212** to limit upward movement of the guide **196** and define its uppermost position relative to various other components. In the embodiment of FIG. **27**, the ledge **197** of the guide **196** is shown (almost) in contact with the lower end **217** of the exemplary coupler **212**. However, other embodiments may include a different mechanism to limit the upward (or downward) movement of the bung plug engagement tool **184** and/or the centering collar **228**. For example, the upper end of the guide **196** may contact one or more transverse surfaces (e.g. the wall defining the upper end of the bore **213**) of the flexible coupler **212**.

(128) For another example of positional flexibility, the centering collar **228** and/or bung plug engagement tool **184** may be provided with angular positional flexibility relative to the drum **30**, such as to allow re-centering and/or proper alignment or seating of the centering collar **228** and/or bung plug engagement tool **184** in imperfect situations (e.g. as described above) and/or for any other purpose(s). In some embodiments, such as shown in FIGS. **17A-18**, the bung plug extractor **50** is configured so that the centering collar **228** can pivot, tilt, or swivel around 360° relative to the vertical axis **220** of the bung plug extractor **50** and/or the drum **30**. In the preferred embodiment, the desired angular positional flexibility of the centering collar **228** is provided by having three independently-acting outer springs **236**. For example, the outer springs **236** may be evenly spaced-apart around the centering collar **228** and the vertical axis **220** of the bung plug extractor **50**. The use of three independently acting (deformable) outer springs **236** may, for example, allow swiveling movement of the centering collar **228** around 360° relative to the vertical axis **220** of the bung plug extractor **50** and/or the drum **30**. If desired, the exemplary bung plug extractor **50** may be configured to allow up to a certain amount (e.g. approximately $\frac{1}{4}$ ") of displacement due to swiveling movement of the centering collar **228** in any direction (around) 360° . In FIG. **29**, for example, the bung opening **32** and lip **35** are approximately $\frac{1}{4}$ " off-center relative to the exemplary bung plug extractor **50** (e.g. vertical axis **220**) and the centering collar **228**. In response to uneven upward forces acting upon the centering collar **228**, one or more of the outer springs **236** deforms to allow the exemplary centering collar **228** (and the associated centering housing **246**, centering cone **266**, floating plate **254**) and bung plug extraction tool **184** to re-center over and align with the bung opening **32** by swiveling, or tilting (downward to the right in the illustration) so that bung plug **34** extraction can proceed. However, other embodiments may allow the centering collar **228** to

move in a range of motion less than 360° and/or more or less than ¼" movement (e.g. ⅛", ½", ¾", 1", etc.) in any or certain directions, or a combination thereof, or not allow relative angular movement of the centering collar **228**.

(129) Referring again to FIGS. **17A-18**, the exemplary outer springs **236** may allow the desired angular positional flexibility of the centering collar **228** in any suitable manner and with any suitable components. In the preferred embodiment, the centering collar **228** is operatively coupled to the floating plate **254**, upon which each of the illustrated outer springs **236** acts. For example, each outer spring **236** may be biased downwardly between a biasing collar **237** (or other component(s) rigidly coupled to its associated spring support rod **238**) and the floating plate **254** (or one or more components (e.g. a flat washer **255**) associated therewith). The exemplary floating plate **254** is able to float (up-and-down and side-to-side) independently on each spring support rod **238** within a limited range of motion between the outer springs **236** and the lower plate **260** (which is rigidly coupled to the rods **238** proximate to the lower ends **238a** thereof). Thus, in the preferred embodiment, if uneven upward forces are placed upon the exemplary centering collar **228**, one or more of the outer springs **236** may react, causing the floating plate **254** and attached centering collar **228** to move up or down, tilt or pivot relative to one or more of the spring support rods **238**, lower plate **269**, vertical axis **220** and/or drum **30** within the floating range of motion of the floating plate **254**. The exemplary bung plug extractor **50** is thus configured to allow relative angular movement of the floating plate **254**, centering collar **228** (and bung plug engagement tool **184**, such as described below) up to ¼" displacement in any direction around the perimeter of the spring support rods **238**.

(130) However, the exemplary outer springs **236** (or other components) may allow the desired relative angular movement of the centering collar **228** in any other manner. Further, other embodiments of the bung plug extractor **50** may include more (four, five, six or more) or less (one or two) outer springs **236**, one or more pairs of nested springs, or any other desired quantity and/or configuration of springs and/or other types of biasing members or other components (e.g. elastic members, swivels, bearings, etc.) to allow the desired relative angular movement or positional flexibility of the centering collar **228** and/or bung plug engagement tool **184** (e.g. more or less displacement (e.g. 1/16", ⅛", ½", ⅝", ¾" or more) in any desired direction and/or within any desired range of motion (e.g. more or less than)) 360°. For example, one or more swivel joints, bearings or other components may also, or instead, allow the desired relative angular movement of the centering collar **228** and/or bung plug engagement tool **184**. However, in some embodiments, relative angular movement of the centering collar **228** and/or bung plug engagement tool **184** may not be provided. Thus, the details, nature, characteristics and operation of the components of the bung plug extractor **50** are not limiting upon the present disclosure and appended claims, except and only to the extent as may be expressly recited and explicitly required in a particular claim hereof and only for such claim(s) and any claims depending therefrom.

(131) Referring now to FIG. **18**, if desired, the bung plug engagement tool **184** may be configured to move in substantial unison with the centering collar **228** in relative angular movement (e.g. as described above) to allow angular positional flexibility of the bung plug engagement tool **184** relative to the drum **30**. Concurrent relative angular movement of the tool **184** and centering collar **228** may be provided in any suitable manner. In the preferred embodiment, the bung plug engagement tool **184** is operatively coupled to the centering collar **228** with respect to relative angular movement. When this feature is included, the bung plug engagement tool **184** may be operatively coupled to the centering collar **228** in any suitable manner. In the preferred embodiment, a centering housing **246** is coupled to and extends at least partially between the floating plate **254** and the centering collar **228**. The centering housing **246**, when included, may have any suitable form, configuration and operation. For example, the illustrated centering housing **246** includes a central bore **248** through which the guide **196** of the bung plug engagement tool **184** extends and is freely slideable up and down. In the preferred embodiment, the central bore **248** of

the exemplary centering housing **246** may be sized so that the guide **196** fits snugly therein (e.g. with minimal or no side-to-side movement of the guide **196** relative to the centering housing **246**) to allow the bung plug engagement tool **184** to move concurrently with the centering housing **246** in swiveling movement and vice versa. If desired, one or more bushing assemblies **252** (or other components) may be associated with the bore **248** and/or centering housing **246** to assist in reducing frictional wear of the guide **196** and/or centering housing **246** as the guide **196** moves up and down relative thereto, for any other purpose(s) or a combination thereof.

(132) It should be noted that the swiveling movement of the centering collar **228** and the bung plug engagement tool **184** may be limited by the flexible coupler **212**, when included, acting on the bung plug engagement tool **184** (e.g. as described above) and/or any other components. In the preferred embodiment, the concurrent swiveling movement of the exemplary centering collar **228** and bung plug engagement tool **184** is limited by both the relevant range of motion of the exemplary floating plate **254** and its connected components and the exemplary universal joint **214**. However, any other configuration of components (e.g. swivel joints) may be provided to allow concurrent swiveling movement of the bung plug engagement tool **184** and the centering collar **228**. Thus, the details of the construction and operation of the components described and shown herein to allow concurrent swiveling movement of the bung plug engagement tool **184** and the centering collar **228** are in no way limiting upon the present disclosure or appended claims, except and only to the extent as may be expressly recited and explicitly required in a particular claim hereof and only for such claim(s) and any claim(s) depending therefrom.

(133) Referring now to FIGS. **30-32**, the centering collar **228** may be rigidly, releasably coupled to the centering housing **246**, such as for removal and replacement of the centering collar **228** due to wear or breakage, to switch out the centering collar **228** to match its inner diameter **230** with the diameter **35c** (e.g. FIG. **28**) of the lip **35** of the particular drum's bung opening **32**, for any other purpose(s) or a combination thereof. In the preferred embodiment, the centering housing **246** includes one or more keys **247** that slideably, releasably, lockingly engage a respective locking groove **232** in the centering collar **228**. However, any other arrangement of parts may be used to releasably couple the centering collar **228** and the centering housing **246**, when included, or they may be non-releasably or non-rigidly coupled together or integrally formed.

(134) Referring back to FIG. **18**, if desired, the centering assembly **180** may include one or more centering cones **266** (or other component(s)) useful to assist in centering the centering collar **228** and bung plug engagement tool **184** relative to the vertical axis **220** of the bung plug extractor **50** and/or the drum **30** at any desired time, for any other purposes or a combination thereof. The centering cone **266**, when included, may have any suitable form, configuration, components and operation. In the preferred embodiment, the centering cone **266** is configured to assist (e.g. along with the outer springs **236**) in realigning or positioning the centering collar **228** and bung plug engagement tool **184** in their “home” positions (e.g. centered on the vertical axis **220** of the bung plug extractor **50** and/or the drum **30**) when the bung plug extractor **50** is not engaged with a drum **30**. For example, the centering cone **266** may be cone-shaped and rigidly releasably coupled (e.g. sandwiched) between the floating plate **254** and the centering housing **246**. Referring to FIG. **32**, if desired, one or more couplers **268** (e.g. screws) may connect the exemplary floating plate **254** (e.g. via holes **258**) to the centering housing **246** (e.g. via holes **251**) and extend through respective passages **270** in the centering cone **266**. In the preferred embodiment, the couplers **268** include three screws. In other embodiments, any desired quantity (e.g. one, two, four, five, six, etc.) and form (e.g. clip, pin, rod, etc.) of couplers **268** or other connection components may be used.

(135) Referring now to FIGS. **18 & 32**, the exemplary centering cone **266** includes a central bore **272** that extends around a neck **249** of the centering housing **246**. In this embodiment, an angled outer circumferential surface **276** of the illustrated centering cone **266** is configured to seat within a complimentary-angled inner diameter **262** of a central bore **264** formed in the lower plate **260** (e.g. in a preferred “home” position). Thus, the exemplary centering cone **266** will typically ensure that

the centering collar **228** and bung plug engagement tool **184** are precisely aligned in their respective home positions, for example, prior to the start of bung plug **34** removal operations. However, the centering cone **266** may have any other shape (non-conical), be constructed of multiple components and operate in any other suitable manner. In some embodiments, the centering cone **266** may not be included, or one or more additional or different components may be used to help center the centering collar **228** and bung plug engagement tool **184** as desired. Thus, the details of the construction and operation of the centering cone **266** and related components as provided herein are in no way limiting upon the present disclosure or appended claims, except and only to the extent as may be expressly recited and explicitly required in a particular claim hereof and only for such claim(s) and any claim(s) depending therefrom.

(136) Referring now to FIGS. **26-28**, the bung plug extractor **50** may be configured to remove the bung plug **34** once it is disconnected from the drum **30**. If this feature is included, any suitable arrangement of components and technique(s) may be used. For a few examples, the bung plug extractor **50** may utilize one or more cups, grippers, claws, tongs, magnets or any other mechanical or other components to physically or otherwise engage or grab the bung plug **34** or allow it to be moved away from the drum **30**. For another example, suction may be used to effectively lock onto or grab the disconnected bung plug **34** and allow it to be moved away from the bung opening **32**.

(137) In the preferred embodiment, the drum manipulation system **10** is configured to provide air suction inside one or more vacuum spaces **280** formed between the mating portion **192** of the bung plug engagement tool **184** and the bung plug **34** (see also FIG. **18**). For example, a vacuum may be provided in the vacuum space **280** through one or more air passageways **284** formed in, or extending at least partially through, the bung plug engagement tool **184** sufficient to lift the disconnected bung plug **34** (e.g. edge **37**) up and at least partially against or proximate to the face **193** of the mating portion **192** (or other part(s)) of the bung plug engagement tool **184** and carry it with the bung plug engagement tool **184** away from the drum **30**. In this manner, the bung plug **34** is “grabbed” or “carried” by the exemplary bung plug engagement tool **184**. However, in other embodiments, suction may be created in a different space or between the bung plug **34** and another component of the bung plug extractor **50** to releasably engage, hold or carry the bung plug **34** for removal, or may not be included. For example, the bung engagement tool **184** of FIG. **21** includes a vacuum cup **274** for grabbing the bung plug **34** (e.g. plastic plug **34b**, FIG. **5**) in various scenarios (e.g. when manipulating a polyethylene drum **30**).

(138) Referring now to FIGS. **18 & 27**, in the preferred embodiment, a vacuum may be provided through the air passageway(s) **284** of the bung plug engagement tool **184** to the vacuum space **280** (or other desired location) in any suitable manner. For example, one or more vacuum sources **288** (e.g. air compressors, vacuum generators, vacuum pumps) may be releasably fluidly coupled (e.g. via one or more hoses **289**) directly to the bung plug engagement tool **184**. For another example, the vacuum source(s) **288** may be coupled (e.g. via one or more hoses **289**) to a vacuum housing **292** which is fluidly coupled to the air passageway(s) **284** of the bung plug engagement tool **184**. In some embodiments, the drum manipulation system **10** may include an internal vacuum source or may be coupled to an external vacuum source. In FIG. **1C**, the vacuum source is a vacuum generator **290** (shown with a vacuum gauge **291**). If desired, the vacuum source(s) **288** and/or other vacuum-related components may be selectively actuated by the controller **12**, but could instead or also be actuated by one or more operators, the driver **204** (e.g. FIG. **17A**), other component(s) of the drum manipulation system **10** or a combination thereof. In some embodiments, the vacuum source may be large capacity (e.g. up to 60 cfm) to pull sufficient volume of air to lift the bung plug **34** (e.g. when a complete fluid tight seal is not formed around the vacuum space **280**). If desired, one or more filters **293** (e.g. FIG. **1C**) may be provided at one or more locations in the vacuum path to prevent metal shavings (e.g. created during rotation of the bung plugs **34**) or other debris from entering the vacuum source(s) **288** or other components, for any other purpose(s) or a combination thereof. In the preferred embodiment, at least one filter **293** (e.g. FIG. **1C**) is provided

on the inlet side of the vacuum source(s) **288** to prevent debris from entering.

(139) Still referring to FIGS. **18** & **27**, when included, the vacuum housing **292** may have any suitable form, configuration, components and operation. In the preferred embodiment, the vacuum housing **292** is a sliding flange, collar, or sleeve, **294** (e.g. FIG. **33**) rigidly releasably coupled (e.g. via snap, or retainer, rings **296**) to the guide **196** of the bung plug engagement tool **184** above the floating plate **254**. The exemplary vacuum housing **292** includes at least one fluid flow passage **300** fluidly coupling the air passageway **284** of the bung plug engagement tool **184** and the vacuum source **288**. If desired, the vacuum housing **292** may be spring-biased downwardly by the inner spring(s) **240** and moveable with the guide **196**. For example, each inner spring **240** may be provided around a respective inner spring rod, or linear bearing shaft, **241** extending through the vacuum housing **292** and anchored, at its lower end **241a**, to the floating plate **254** (see e.g. FIGS. **30-31**). While three inner springs **240** are shown, any other quantity (one, two, four or more) of inner springs **240** (or other biasing components) may be included. The exemplary vacuum housing **292** and bung engagement tool **184** are thus concurrently moveable up and down relative to the inner spring rods **241**, floating plate **254**, lower plate **260**, centering cone **266**, centering housing **246**, centering collar **228** and drum **30**. However, the vacuum housing **292** may have any other configuration and may be fluidly coupled between one or more vacuum sources **288** and the bung plug engagement tool **184** and/or moveable relative to any combination of components in any other suitable manner (e.g. without the use of inner springs **240**) or not at all. Thus, the details of the construction and operation of the vacuum housing **292** and other components and techniques described and shown herein to provide a vacuum for lifting or otherwise engaging the bung plug **34** are in no way limiting upon the present disclosure or appended claims, except and only to the extent as may be expressly recited and explicitly required in a particular claim hereof and only for such claim(s) and any claim(s) depending therefrom.

(140) Referring now to FIGS. **17A** & **28**, once the disconnected bung plug **34** has been grabbed by the exemplary bung plug engagement tool **184** or other component (e.g. the vacuum has been actuated and the bung plug **34** sucked up to the exemplary bung plug engagement tool **184**), the bung plug extractor **50** and bung plug **34** may be selectively moved away from the drum **30**. Before shuttling away the entire bung plug extractor **50**, it may be desirable to initially retract, or move the bung plug **34** (e.g. straight up) away from the drum **30** and clear of the bung opening **32** to avoid knocking or tapping the bung plug **34** against the lip **35**, or receiving threads **33** of the bung opening **32**, or other part of the drum **30**. Such event could, for example, release the suction of the vacuum (when included) and/or otherwise cause the bung plug **34** to drop from the bung plug engagement tool **184**, interrupt or inhibit further operations, or cause other undesirable consequences. In the preferred embodiment, the drum manipulation system **10** is configured to initially retract the vacuum housing **292** and bung plug engagement tool **184** up (e.g. approximately 1.0", or more or less) away from the drum **30** against the downward biasing forces of the inner springs **240** before shuttling the bung plug extractor **50** away from the drum **30**. For example, the vacuum housing **292** and bung plug engagement tool **184** may be initially moved up by a bung plug retraction actuator **308**.

(141) When included, the bung plug retraction actuator **308** may have any suitable form, configuration, components and operation sufficient to raise-up the vacuum housing **292**, bung plug engagement tool **184** (and/or other components) and bung plug **34** clear of the bung opening **32** and drum **30**. For example, referring to FIGS. **28** & **31**, the actuator **308** may be a pneumatic or hydraulic cylinder **310** having an up-and-down moving piston **312** associated with the vacuum housing **292** and/or bung plug engagement tool **184** and configured to move it/them up as desired. In the preferred embodiment, a bracket **314** (e.g. L-bracket) extends outwardly from the piston **312** and is positioned to engage the vacuum housing **292** and/or bung plug engagement tool **184** (see also FIG. **32**). For example, the bracket **314** may have a fork-shaped, U-shaped or similarly-shaped base **316** that at least partially straddles, or extends around, the guide **196** of the bung plug

engagement tool **184** below the vacuum housing **292** (e.g. FIG. 17B). The illustrated bracket **314** is rigidly releasably coupled to the piston **312**, but could instead be integrally formed therewith, or non-rigidly or non-releasably coupled thereto. The exemplary bracket **314** is selectively moveable up by the piston **312** to drag the exemplary vacuum housing **292** (bung plug engagement tool **184** and bung plug **34**) up the desired distance to clear the disconnected bung plug **34** away from the drum **30**.

(142) Referring to FIGS. 27 & 31, if desired, a “start” or “home” position of the exemplary piston **312** and connected bracket **314** may be a down position relative to the bung plug engagement tool **184** so that the bracket **314** is out of the way of the bung plug engagement tool **184** and will not interfere with the action of the bung plug engagement tool **184** during disengagement of the bung plug **34** from the drum **30** and/or for any other purposes. After the bung plug **34** has been disconnected and grabbed by the exemplary bung plug engagement tool **184**, the illustrated piston **312** may be selectively moveable up such as described above and thereafter selectively movable down. In the preferred embodiment, the actuator **310** is anchored to the side of the floating plate **254**, such as via a connection plate **318** (e.g. FIG. 32), but could be mounted elsewhere as desired. Moreover, the bung plug retraction actuator **308** may have any other form, components and operation to selectively move the bung plug engagement tool **184** and bung plug **34** away from the drum **30** as desired. In some embodiments, particularly those not utilizing a vacuum housing **292**, the bung plug retraction actuator **308** may be configured to directly move the bung plug engagement tool **184** and/or bung plug **34** away from the drum **30** (e.g. without the use of a vacuum housing **292**), or may not be included. Moreover, any other suitable components and techniques (e.g. mateable engagement, gripping engagement, etc.) may be used to selectively move the bung plug **34** away from the drum **30**. Thus, the details of the construction and operation of the components and techniques described and shown herein to initially move the engaged bung plug **34** away from the drum **30** are in no way limiting upon the present disclosure or appended claims, except and only to the extent as may be expressly recited and explicitly required in a particular claim hereof and only for such claim(s) and any claim(s) depending therefrom. Further, the bung plug extractor **50** may have any other suitable form configuration, components and operation. For example, the bung plug extractor **50** may incorporate or use any part, or all of, the apparatus and techniques disclosed in German Patent Pub. No. DE3715795A1 to Feige published on Nov. 24, 1988 and German Patent Pub. No. DE3715795C2 to Feige published on Nov. 22, 1990, both from German Patent App No. DE19873715795 filed on May 12, 1987 and entitled “Screwing Device for the Screw-in Lids of Barrel Bungholes”, all the details and contents of which are hereby incorporated by reference herein in their entireties.

(143) Referring back to FIGS. 1A-2, after the bung plug **34** has been disconnected from the drum **30** by the exemplary bung plug extractor **50** (and, if desired, initially cleared of the bung opening **32**), the bung plug extractor **50** and/or bung plug **34** may be moved away from the drum **30**, or the drum **30** moved away therefrom. In some embodiments, a separate mechanism may remove the bung plug **34** from the bung plug extractor **50**. In the preferred embodiment, the bung plug extractor **50** carries the bung plug **34** away from the drum **30**. For example, the bung plug **34** may be carried away from the drum **30** by the bung plug extractor **50** (e.g. by maintaining suction) as the bung plug extractor **50** is shuttled, or retracted, away from the drum **30**. In such instances, the bung plug extractor **50** may be selectively moveable away from the drum **30** in any suitable manner. For example, one or more drivers **46** may be operatively coupled to the bung plug extractor **50** and, if desired, electronically coupled to electronic controller **12**, for selectively moving the bung plug extractor **50** in one or more linear, arc-like and/or angular paths back and forth between positions, similarly as described above. In the preferred embodiment, the bung plug extractor **50** is typically moveable between operating and home positions by the second driver **46b** in a vertical path (up clear of the drum **30**), then by the first driver **46a** in a linear path sideways (e.g. horizontally) away from the drum **30**. However, in other embodiments, the bung plug

extractor **50** may be moveable away from the drum **30** in any other desired manner, along a single path or one or more non-linear paths (e.g. an arc-like motion, angular motion, etc.), with the use of any other form or combination of driver(s) or other components controlled by the controller **12**, other component(s) (e.g. sensor(s)), one or more operators or a combination thereof.

(144) Referring to FIGS. **1A** & **34**, during retraction of the exemplary bung plug extractor **50**, the bung plug **34** may, if desired, be moved to a desired location for storage, disposal, reuse or any other purpose, or may be retained by the bung plug extractor **50** or other component(s). For example, the bung plug **34** may be moved to a location when it can be re-engaged with the same (or a different) drum **30**. In the preferred embodiment, the bung plug extractor **50** is configured to be selectively actuated to drop, or place, the bung plug **34** on a bung plug conveyor **328**. For example, after moving the bung plug extractor **50** up, the bung plug extractor **50** may be moved initially in a rearward horizontal path (toward the rear side **106** of the drum manipulation system **10**) away from the drum **30** to a position over the bung plug conveyor **328** to allow the bung plug **34** to be released onto the bung plug conveyor **328**. In some embodiments, the bung plug extractor **50** may then be moved down to place the bung plug **34** on the bung plug conveyor **328**. For example, once the exemplary bung plug extractor **50** is in a down position over the bung plug conveyor **328** (e.g. with the bung plug **34** resting on the bung plug conveyor **328**), the vacuum may be de-actuated to release suction, or reversed to provide air pressure out of the bung plug engagement tool **184**, to ensure timely release of the bung plug **34**, hold the bung plug **34** in position on the bung plug conveyor **328** as the bung plug extractor **50** is then lifted up away from the bung plug conveyor **328** and drum **30** (e.g. to a home position), for any other suitable purpose(s) or a combination thereof. After clearing the bung plug **34** and bung plug conveyor **328**, the vacuum may be turned off. However, any other technique or sequence may be used.

(145) Still referring to FIGS. **1A** & **34**, the exemplary bung plug conveyor **328** is configured to convey the bung plug **34** to a location where it can later be picked up and re-engaged with the same drum **30**. For example, the illustrated bung plug conveyor **328** may be configured to convey the bung plug **34** to a position adjacent to a third (e.g. bung plug installation) station **122** of the drum manipulation system **10** (or drum filing system **70**). However, the bung plug conveyor **328** may move the extracted bung plugs **34** to a different destination. Moreover, the extracted bung plug **34** may instead be deposited onto, or into, a different type of device (e.g. vibratory bung plug feeder, disposal bin) for reuse or disposal or any other purpose.

(146) When included, the bung plug conveyor **328** may have suitable form, configuration, components and operation. For example, the bung plug conveyor **328** may include a mini-mover belt **334** (e.g. 2.9" wide or more or less) configured to be selectively actuated by any suitable driver **332**, such as an electric motor. If desired, the driver **332** may be electronically coupled to the controller **12** or other component of the drum manipulation system **10** for selective control thereof. In the preferred embodiment, the exemplary bung plug conveyor **328** is formed with a length suitable to transport the removed bung plugs **34** one-at-a-time in a linear path from the first station **118** to the third station **122** (e.g. 40", or more or less) and positioned rearward of (e.g. and parallel to) the drum conveyor **88** proximate to the rear side **106** of the drum manipulation system **10**. One or more sensors **336** (e.g. photo-eye sensors) may be used to detect the presence of a bung plug **34** at each desired location on the bung plug conveyor **328**. However, the bung plug conveyor **328** may have any other configuration, location and operation and the bung plugs **34** may be deposited thereon (and/or removed therefrom, such as described below) in any other suitable sequence or manner. Thus, the details of the construction and operation of the bung plug conveyor **328** and other components and techniques described and shown herein relating to the disposition of the removed bung plug **34** are in no way limiting upon the present disclosure or appended claims, except and only to the extent as may be expressly recited and explicitly required in a particular claim hereof and only for such claim(s) and any claim(s) depending therefrom.

(147) Referring to FIGS. **1A** & **28**, if desired, the bung plug **34** may be placed on the exemplary

bung plug conveyor **328** in a particular orientation and/or with the bung plug engagement tool **184** in a particular position so that the exemplary bung plug installer **60** (or other component) can later engage or pick up the bung plug **34**, for any other purpose(s) or a combination thereof. For example, the tabs **194** of the mating portion **192** of the illustrated engagement tool **184** of the bung plug extractor **50** (e.g. FIG. **28**) and bung plug installer **60** (e.g. FIG. **39**) may be specifically oriented at or near the same positions when the bung plug **34** is placed on and removed from the bung plug conveyor **328**, respectively. In embodiments of bung plug engagement tool **184** having first and second tabs **194**, for one example, the tabs **194** may be positioned at, or near, respective three o'clock and nine o'clock (or any other desired) positions relative to the front side **104** of the drum manipulation system **10** for plug drop-off and pick-up. Similarly, or alternately, when a bung plug **34** having protrusions **38** is used, the bung plug **34** (e.g. FIGS. **4**, **5**) may be placed on the bung plug conveyor **328** with the protrusions **38** in a particular orientation. For one example, in situations using bung plugs **34** having first and second protrusions **38** (e.g. bung plug **34a**, FIG. **4**), the protrusions **38** may be specifically oriented at, or near, respective six o'clock and twelve o'clock (or any other desired) positions relative to the front side **104** of the drum manipulation system **10** when the bung plug **34** is placed on and removed from the bung plug conveyor **328**. This would allow the bung plug engagement tool **184** of the exemplary bung plug installer **60** (e.g. FIG. **39**), as described further below, to later be oriented in approximately the same position as the bung plug engagement tool **184** of the bung plug extractor **50**, such as to efficiently and effectively pick up the bung plug **34**, avoid the exemplary tabs **194** (or other part of the bung plug installer **60**) from undesirably hitting the protrusions **38** (or other part of the bung plug **34**) and/or preventing the creation of sufficient suction to pick up the bung plug **34** or otherwise hinder the ability of the bung plug installer **60** to grab the bung plug **34** and/or for any other purpose(s). However, any other orientation of the protrusions **38** or other part(s) of the bung plug **34** and/or the tabs **194** (or other part(s)) of the respective bung plug engagement tools **184** of the bung plug extractor **50** and bung plug installer **60** or techniques may be used to selectively position the bung plug **34** on the bung plug conveyor **328** or otherwise assist the bung plug installer **60** in picking up the plug **34** from the bung plug conveyor **328**, when this capability is included.

(148) Referring back to FIGS. **1A-2**, after the bung plug **34** is removed from a “first” drum **30** (e.g. and the bung plug extractor **50** and bung plug **34** are moved away), the first drum **30** may be further manipulated as desired (e.g. inspected, filled, cleaned, modified, etc.). In the preferred embodiment, the first drum **30** is moved to the second station **120**. For example, retraction of the bung plug extractor **50** away from the first drum **30** or other event may cause the drum conveyor **88** to move it to the second station **120** (e.g. at the direction of the controller **12**, other component(s) or operator(s)). As the first drum **30** is transported away from the exemplary first station **118**, one or more sensors **93** or other component(s) may be configured to detect the arrival of the first drum **30** at a particular position along the length of the drum conveyor **88** that represents the second station **120** and cause the drum conveyor **88** to stop, or otherwise deposit the first drum **30** at that location. In the preferred embodiment, upon detection of the first drum **30** at the desired position, one or more sensors **93** or other component(s) signal the electronic controller **12**, which signals the drum conveyor **88** to stop moving. However, in other embodiments, the sensor(s) **93**, or other component(s), may directly cause the drum conveyor **88** to stop or may not be included. Some embodiments may not involve movement of the drum **30** between multiple stations. For example, all of the drum manipulation activities may occur at only one station.

(149) Still referring to FIGS. **1A-2**, if desired, a second drum **30** may be deposited or provided on the drum conveyor **88** behind the first drum (e.g. at or near the entrance end **100** of the drum manipulation system **10** and/or drum filling system **70**), so that movement of the first drum **30** to the exemplary second station **120** will concurrently move the second drum **30** to the first station **118** (e.g. for removal of the bung plug **34** therefrom). Thereafter, the desired drum manipulation operations (if any) at the respective stations **118**, **120** may be performed to both the first and second

drums **30**. A third drum **30** may be similarly added and moved to the first station **118** as the first drum **30** moves to the third station **122** and the second drum moves to the second station **120** for drum manipulations operations there (if any), then a fourth drum **30**, and so on. In the preferred embodiment, any desired quantity of drums **30** may be subsequently added and moved in unison (spaced apart appropriately) to each successive station, then out of the drum manipulation system **10** (and drum filling system **70**) at the exit end **102** in an “assembly-line” style of continuous processing.

(150) In the illustrated drum manipulation system **10** (and/or drum filling system **70**), the second station **120** involves the filling of the drum **30** with liquid by the drum filler **74**. However, any other activity or no activity may take place at the second station **120**. After the desired activity or inactivity at the second station **120**, or otherwise whenever desired, the exemplary bung plug installer **60** may be used to attach a bung plug **34** to the bung opening **32** of the subject drum **30**. In this embodiment, the drum **30** is moved to the third station **122** (e.g. similarly as described above with respect to movement of the drum **30** from the first station **118** to the second station **120**) for installation of a bung plug **34**. However, in other embodiments, the bung plug installer **60** may attach a bung plug **34** to the drum **30** at the second station **120** (the first station, any other station or location).

(151) Referring now to FIGS. **35** & **36**, the bung plug installer **60** may have any suitable form, configuration, components and operation. For example, the bung plug extractor **50**, such as described above, may be used as the bung plug installer **60**. In other embodiments, the bung plug installer **60** may be separate and distinct from the bung plug extractor **50**, but have most of the same components, features, details and operation as the bung plug extractor **50** (such as described above). For example, the bung plug installer **60** may include the same centering assembly **180** (e.g. centering collar **228**, centering housing **246** and centering cone **266**), bung plug engagement tool **184**, carrier **200**, vacuum housing **292** and related components as described above with respect to the exemplary bung plug extractor **50**, the descriptions above of which are hereby incorporated by reference herein in their entireties. In such instance, these exemplary components may be used generally in reverse by the illustrated bung plug installer **60** to grab (e.g. suck up) the bung plug **34** off the bung plug conveyor **328** (e.g. FIG. **1A**) or other location, center the bung plug engagement tool **184** and bung plug **34** over the bung opening **32** and install the bung plug **34** (carried thereby) into the bung opening **32** (e.g. rotating the bung plug engagement tool **184** in the opposite direction as during bung plug extraction by the bung plug extractor **50**). In the preferred embodiment, some components of the bung plug installer **60** may be modified or have modified capabilities as those of the bung plug extractor **50**. For example, the driver **204** (for actuating the guide **196** of the bung plug engagement tool **184**) of the bung plug installer **60** may include the same and additional capabilities as the driver **204** of the bung plug extractor **50**. For example, the driver **204** of the bung plug installer **60** may be a servo-motor **350** having a selectively rotatable shaft operatively coupled (e.g. via a gear box **354**) to the guide **196** of the bung plug engagement tool **184** for rotation thereof in the same manner, or similarly, as described above and shown in the appended drawings as the air motor **206** (e.g. FIG. **17C**) of the bung plug extractor **50**, plus additional components and capabilities (such as described below). Furthermore, various of the features of the exemplary bung plug extractor **50** may not be necessary in the exemplary bung plug installer **60**, such as the bung plug retraction actuator **308** (e.g. FIG. **31**) and related components (e.g. bracket **314**, connection plate **318**), and thus may not be included in the bung plug installer **60**.

(152) Referring to FIGS. **1A-B** & **2**, in the preferred embodiment, once the drum **30** is positioned as desired for installing a bung plug **34** in the drum **30** (e.g. moved to the third station **122**), the exemplary bung plug installer **60** may be shuttled, or moved, into position relative to the drum **30** and/or the drum **30** may be moved in a desired position relative to the bung plug installer **60**. For example, the bung plug installer **60** may be moved from at least one “home” position (e.g. away from the drum **30**, similar to the home positions of the exemplary bung locator **40** and bung plug

extractor **50** as described above) to an “operating” position over and/or in contact with the drum **30**. In at least one exemplary home position, the bung plug installer **60** may be sufficiently clear of the drum **30** to allow other one or more components (e.g. dust cap crimper **98**) to engage or perform other drum manipulation operations on the drum **30**, allow the drum **30** to be moved to a different location, for any other desired purpose(s) or a combination thereof. In at least one operating position, the exemplary bung plug installer **60** may be over the bung opening **32** of the drum **30**. In the preferred embodiment, for example, FIG. **36** illustrates the exemplary bung plug installer **60** over the drum **30** prior to being lowered down onto the drum **30**. FIG. **37** illustrates the bung plug installer **60** of the preferred embodiment installing the bung plug **34** in the drum **30** and FIG. **39** illustrates the exemplary bung plug installer **60** after installing the bung plug **34** and moving up away from the drum **30**. (It should be noted that FIG. **39** also represents an exemplary position of the bung plug installer **60** of the preferred embodiment prior to picking up a bung plug **34** prior to installation.)

(153) Referring again to FIGS. **1A-B** & **2**, in the preferred embodiment, the bung plug installer **60** may be selectively moveable relative to the drum **30** as desired in any suitable manner. For example, one or more drivers **46** may be operatively coupled to the bung plug installer **60** and, if desired, electronically coupled to electronic controller **12**, for selectively moving the bung plug installer **60** in one or more linear, arc-like and/or angular paths back and forth between positions, similarly as described above with respect to the movement of the exemplary bung locator **40**. In the preferred embodiment, the bung plug installer **60** is moveable between home and operating positions by a third driver **46c** (e.g. linear pneumatic or hydraulic cylinder) in a first linear path sideways (e.g. horizontally) toward the drum **30** and then by a fourth driver **46d** (e.g. linear pneumatic or hydraulic cylinder) in a linear vertical path down to the drum **30**. If desired, one or more sensors **356** (e.g. proximity sensors **358**, FIG. **40**) may be used to assist in positioning the bung plug installer **60** for each of its positions. In other embodiments, the bung plug installer **60** may be moveable toward and away from the drum **30** in any other desired manner, along a single path or one or more non-linear paths (e.g. an arc-like motion, angular motion, etc.), with the use of any other form or combination of driver(s) or other components controlled by the controller **12**, other component(s), one or more operators or a combination thereof.

(154) For installing a bung plug **34** in the drum **30**, the bung plug **34** should preferably be brought into position on, over or in the bung opening **32** for engagement therewith by the exemplary bung plug installer **60**. The bung plug **34** may be positioned on, over or in the bung opening **32** in any suitable manner. For example, a robotic arm, or other component, may place the bung plug **34** on, over or in the bung opening **32** (e.g. prior to engagement by the bung plug installer **60**). For another example, a robotic arm, or other component, may deliver the bung plug **34** to the bung plug installer **60**, which will then position the bung plug **34** on, over or in the bung opening **32**. For yet another example, the bung plug installer **60** may itself store one or more bung plugs **34** for one-by-one successive positioning onto the respective bung openings **32** of each successive drum **30** (e.g. with a built-in automatic or associated dispenser).

(155) Referring still to FIGS. **1A-B** & **2**, in many embodiments, the bung plug installer **60** itself retrieves the bung plug **34** and brings it to the bung opening **32** of the subject drum **30**. In such instances, the installer **60** may use any suitable configuration of components to retrieve and carry the bung plug **34**. Similarly, as mentioned above for the bung plug extractor **50**, the bung plug installer **60** may utilize one or more cups, grippers, claws, tongs, magnets or any other mechanical or other components to retrieve (physically or otherwise engage or grab) the bung plug **34** (at any desired bung plug storage location or waiting area). For another example, suction may be used to effectively lock onto, or grab, the bung plug **34** and carry it to the bung opening **32**. Referring to FIG. **36**, preferably, the drum manipulation system **10** is configured to provide air suction inside one or more vacuum spaces **280** formed between the mating portion **192** of the bung plug engagement tool **184** of the bung plug installer **60** and the bung plug **34**, in the same manner, or

similarly, as described above with respect to the bung plug extractor **50** (e.g. FIGS. **18**, **26-28**). Thus, the details, components and operations provided above and shown in the appended drawings relating to use of a vacuum of the exemplary bung plug extractor **50** are hereby incorporated by referenced herein in their entirety with respect to the exemplary bung plug installer **60**. For example, a vacuum may be formed in the vacuum space **280** through one or more air passageways **284** formed in, or extending at least partially through, the bung plug engagement tool **184** of the bung plug installer **60** sufficient to lift the bung plug **34** (e.g. edge **37**, FIG. **4**) up and at least partially against or proximate to the face **193** (or other portion(s)) of the mating portion **192** (e.g. FIG. **39**) of the bung plug engagement tool **184** and carry it with the bung plug engagement tool **184**. However, in other embodiments, suction may be created in a different space or between the bung plug **34** and another component of the bung plug extractor **50** to releasably engage, hold or carry the plug **34** for removal, or may not be included. For example, the bung engagement tool **184** of FIG. **21** (as described above) may be used in the bung plug installer **60**.

(156) Referring back to FIGS. **1A** & **2**, when the exemplary bung plug installer **60** moves to the drum **30** and retrieves the bung plug **34** from a bung plug conveyor **328** (or other plug depository), the bung plug installer **60** may be moved (e.g. from a home position) to the bung plug conveyor **328** to allow the bung plug engagement tool **184** bung plug installer **60** (e.g. FIG. **36**) to suck up (or otherwise grab or engage) the bung plug **34** and carry it (e.g. by maintaining suction) to the drum **30**. This may be done in any suitable manner and with any desired components. In the preferred embodiment, the bung plug installer **60** is moved in a horizontal path toward the bung plug conveyor **328** via the third driver **46c** to a “first intermediate” position to pick up the bung plug **34**. For example, one or more sensors **356** (e.g. FIG. **40**), such as a proximity sensor **358**, may detect when the bung plug installer **60** has arrived at the desired first intermediate position and signal the controller **12** (other component or operator). The exemplary controller **12** (other component or operator) may then cause the third driver **46c** to stop moving and initiate one or more other actions (e.g. direct the bung plug installer **60** to pick up a bung plug **34**). In other embodiments, the sensor(s) **356** may communicate directly with the driver **46c**, other component(s), an operator or combination thereof.

(157) Referring to FIGS. **1A** & **39**, if it is desired to orient the bung plug engagement tool **184** of the bung plug installer **60** in a particular manner to pick up the bung plug **34** (e.g. from the bung plug conveyor **328** or other location), the bung plug engagement tool **184** is so oriented in any suitable manner and at any desired time. In the preferred embodiment, before lowering the bung plug engagement tool **184** of the bung plug installer **60** to pick up the bung plug **34**, the tabs **194** of the mating portion **192** of the bung plug engagement tool **184** (e.g. FIG. **20**) may be oriented over spaces between protrusions **38** of the bung plug **34** (e.g. FIGS. **4-5**) by rotating the mating portion **192**. For one example, if the bung plug **34a** (FIG. **4**) was placed on the bung plug conveyor **328** (or other plug depository) with its first and second protrusions **38** oriented at, or near, the six and twelve o'clock positions, the tabs **194** of the mating portion **192** of the bung plug engagement tool **184** may be rotated to, or near, the three and nine o'clock positions. Of course, these particular positions are provided herein for exemplary purposes only and any other desired positions may be used. Alternately, the respective tabs **194** of the bung plug engagement tool **184** of the exemplary bung plug installer **60** may be placed in the same approximate or exact (known) positions in which the respective tabs **194** of the bung plug extractor **50** were situated when the bung plug **34** was placed on the bung plug conveyor **328**, such as described above.

(158) In the present embodiment, the bung plug engagement tool **184** of the bung plug installer **60** is rotated while moving toward the bung plug conveyor **328** (or other bung plug **34** depository). However, in other embodiments, the bung plug engagement tool **184** may be rotated or otherwise oriented at another time, such as before the bung plug installer **60** is moved to the location of the bung plug **34** or after it arrives. The exemplary drum manipulation system **10** is configured so that the same bung plug **34** that was removed from a particular drum **30** will typically be retrieved from

the bung plug conveyor **328** and re-engaged with the same drum **30**. Reinstalling the same bung plug **34** that was removed from the drum **30** may have advantages, such as, for example, providing greater assurance that the bung plug **34** will properly fit and mate with the drum **30**. However, this capability is not required.

(159) Referring to FIGS. **35** & **36**, in the preferred embodiment, the bung plug engagement tool **184** of the bung plug installer **60** is rotated into the desired orientation over the bung plug **34** (before picking it up) with the use of one or more alignment sensors **360** coupled to, or extending from, a non-rotating part of the bung plug installer **60** and configured to detect the presence of one or more flags **368** (having a particular pre-set position) on one of the rotating parts of the bung plug installer **60** during rotation of the bung plug engagement tool **184** over the bung plug **34**. The alignment sensor(s) **360** and flag(s) **368** may have any suitable form, configuration, location and operation. For example, the alignment sensor **360** may be a proximity sensor **362** carried on a mounting plate **371** or other component affixed to the bung plug installer **60**. The exemplary flag **368** may include one or more reflectors, pieces of key stock or other items welded or otherwise affixed to, or integral with, the flexible coupler **212** (e.g. the upper link **215** of the universal joint **214**) or other moving component of the bung plug installer **60** or a marking, irregularity or geometric feature (e.g. protrusion, indentation) thereon. However, the alignment sensor(s) **360** and flag(s) **368** may have any other suitable form and configuration and be provided on any other suitable components of the drum manipulation system **10**. For example, the alignment sensor(s) **360** may instead be coupled to, or provided on, the driver **204**, upper plate **239** or another component coupled thereto, and the flag(s) **368** coupled to, or provided on, a different part of the flexible coupler **212**, the shaft **208** of the driver **204**, the guide **196** or another rotating part of the bung plug engagement tool **184** of the bung plug installer **60**.

(160) Still referring to FIGS. **35** & **36**, in the preferred embodiment, when the exemplary flag **368** is aligned under the alignment sensor **360**, the bung plug engagement tool **184** is in the desired orientation to move down and engage the bung plug **34**. At that time, for example, the exemplary alignment sensor **360** may be configured to signal another component (e.g. the driver **204** and/or controller **12**) to stop rotation of the bung plug engagement tool **184** or take other action or may direct such action itself. However, any other arrangement of parts and techniques may be used to orient the bung plug engagement tool **184** of the bung plug installer **60** for engaging the bung plug **34**. Thus, the components and techniques described and shown herein for orienting the bung plug **34** or bung plug engagement tool **184** to pick up the bung plug **34** are in no way limiting upon the present disclosure or appended claims, except and only to the extent as may be expressly recited and explicitly required in a particular claim hereof and only for such claim(s) and any claim(s) depending therefrom.

(161) Referring to FIGS. **1A** & **39**, after the exemplary bung plug engagement tool **184** of the bung plug installer **60** is oriented as desired, it may be moved down to pick up the bung plug **34** in any suitable manner. In the preferred embodiment, the bung plug installer **60** is moved in a vertical path down to a “second intermediate” position via the fourth driver **46d** until the bung plug engagement tool **184** engages the bung plug **34** as desired. For example, when a vacuum is used to engage the bung plug **34** (e.g. to create suction in the vacuum space **280**), the second intermediate position may be achieved when the face **193** of the mating portion **192** of the bung plug engagement tool **184** abuts the edge **37** of the bung plug **34**, or the mating portion **192** is close enough to the bung plug **34** to allow sufficient vacuum suction to suck up or “grab” the bung plug **34**. If desired, the vacuum source **288** (or other component) may be configured to indicate whether sufficient vacuum suction has been created to lift or grab the bung plug **34**, or otherwise if the bung plug **34** has been lifted, and signal the controller **12** or other component (e.g. to initiate the next action(s)). If it is determined that sufficient vacuum suction is not detected (or the bung plug **34** has not been engaged), the drum manipulation system **10** may be configured to lift the exemplary bung plug installer **60**, rotate the bung plug engagement tool **184** a desired amount (e.g. 10° or more or less)

and try again to engage the bung plug **34**. This process may be repeated a desired number of times (three or more or less) in an effort to grab the bung plug **34** before the controller **12** (or other component) reports a fault event and/or issues a fault signal. However, any other components and techniques may be used to pick up the bung plug **34**. For example, when a vacuum is not used to engage the bung plug **34**, the bung plug engagement tool **184** may engage the bung plug **34** in another suitable manner (such as mentioned above) and, if desired, any suitable components may be used to determine whether successful engagement is achieved.

(162) Referring to FIGS. **1A** & **36**, in the preferred embodiment, after the exemplary bung plug **34** has been grabbed by the bung plug engagement tool **184**, the bung plug installer **60** is moved to an operating position over the bung opening **32** of the drum **30**. Any sequence and components may be used to move the bung plug installer **60** to an operating position. For example, the fourth driver **46d** may be actuated to move the bung plug installer **60** up, the third driver **46c** actuated to move the bung plug installer **60** forward (toward the front side **104** of the drum manipulation system **10**) and the fourth driver **46d** actuated again to move the bung plug installer **60** down to an operating position, all while maintaining engagement with the bung plug **34** (e.g. maintaining vacuum suction). After the exemplary bung plug installer **60** arrives at the operating position (e.g. FIGS. **41-44**), the centering assembly **180**, bung plug engagement tool **184** and other components of the exemplary bung plug installer **60** may be configured to operate in substantially the same manner as described above and shown in the appended drawings with respect to the bung plug extractor **50**, but, in this instance, to install the bung plug **34** into the bung opening **32** (e.g. FIGS. **36-39**).

(163) Referring now to FIGS. **36** & **37**, in the preferred embodiment, since the bung plug **34** threadably engages the bung opening **32**, the bung plug **34** is ideally positioned over the bung opening **32** so that the lowermost of its threads **39** aligns over or abuts the uppermost thread(s) **33** of the opening **32**. Thereafter, the mating portion **192** of the exemplary bung plug engagement tool **184** is rotated (e.g. clockwise) to engage the bung plug **34** in the hole **32**. If desired, the bung plug installer **60** may be capable of registering, or properly aligning, the threads **39** of the bung plug **34** with the threads **33** of the bung opening **32** to ensure proper alignment for threadable installation, provide improved accuracy, efficiency and effectiveness of bung plug installation operations, for any other reason(s) or a combination thereof. The threads **39**, **33** may be registered or properly aligned in any suitable manner and with any suitable components.

(164) In some embodiments and circumstances, it may be desirable to uninstall the bung plug **34** from the drum **30** and try to install it again (e.g. if the bung plug **34** does not properly engage the bung opening **32**). In such instances, the exemplary bung plug engagement tool **184** of the bung plug installer **60** is rotated in the opposite direction (e.g. counterclockwise) to unscrew the bung plug **34** from the drum **30**. Thus, the bung plug installer **60** of the preferred embodiment may also be used to uninstall (and in some cases to extract) the bung plug **34** from the drum **30**.

(165) Referring to FIGS. **36-38**, the exemplary bung plug installer **60** may include additional capabilities, such as the ability to (i) measure the amount of vertical movement of the bung plug **34** in the bung opening **32** during installation, (ii) count the number of revolutions of rotation (e.g. or fractions of a revolution) of the bung plug **34** as it threadably engages the drum **30** (when applicable), (iii) measure the amount of torque applied to the bung plug **34** during installation (when applicable), any other capabilities, any variations thereof or a combination thereof. As used herein and in the appended claims the terms “bung plug insertion variables” and variations thereof mean and refer to any one or more of the above variables (variables measured in items (i)-(iii), other variables relating to installation of the bung plug **34** and variations thereof). Different embodiments may include measuring any one or more (or none) of the bung plug insertion variables, while some embodiments may not measure any bung plug insertion variables.

(166) The bung plug insertion variable(s) may be useful, for example, to assist in preventing damage to the bung plug **34** and/or drum **30** during installation of the bung plug **34** (e.g. stripping or cross-threading threads of the bung plug **34** and/or bung opening **32**), determining when to stop

rotating the bung plug **34**, ensuring the bung plug **34** is properly engaged with the drum **30** (e.g. without excessive torque applied to the bung plug **34**, not inserted too deep or too shallow in the drum **30** or after too few or too many rotations of revolution), prevent a dust-cap from being placed on a drum **30** with a cross-threaded or damaged bung plug **34** (e.g. FIG. **38**), satisfying regulatory criteria or quality control standards and/or reporting requirements or drum certifications (e.g. DOT, Six Sigma), identifying drums **30** that do not meet particular criteria or standards, providing data to report any of the same, for any other purpose(s) or a combination thereof. In some embodiments, one or more bung plug insertion variables may be measured or determined in real-time on a continuing or as-desired basis. Based upon the measurement(s) taken of one or more bung plug insertion variables, the exemplary drum manipulation system **10** (e.g. controller **12**) is preferably configured to make one or more determinations (e.g. and initiate, or signal for, one or more responsive actions when deemed necessary) in real-time on an on-going basis, such as whether the torque applied to the bung plug **34** after a particular length of downward vertical movement of the bung plug **34** and/or number of revolutions of rotation of the bung plug **34** meets or exceeds a desired torque value, the bung plug **34** has been cross-threaded, other determination or a combination thereof.

(167) Still referring to FIGS. **36-38**, if desired, the drum manipulation system **10** (e.g. controller **12**) may be configured to provide (e.g. electronically transmit) bung plug insertion variable data (e.g. in real-time) to an operator, with other entities, such as customers, regulatory authorities and the like and/or to any other location. In some circumstances, for example, the controller **12** can determine if a desired torque value applied to the bung plug **34** is achieved before a desired amount of downward vertical movement of the bung plug **34** and/or number of revolutions of rotation of the bung plug **34**. When this occurs, for example, the controller **12** may provide a default (or other) signal, initiate one or more actions to be taken (e.g. remove the drum **30** from the drum manipulation system **10**), report pertinent data to a desired destination or a combination thereof. For another example, the controller **12** may be configured to determine (e.g. based upon torque, rotations and vertical travel) if the bung plug **34** was cross-threaded when installed in the drum **30** and signal an operator (via screen display, default signal, etc.) that such drum **30** should not be further processed (e.g. have a dust cap installed) and allow the drum **30** to be removed from the drum manipulation system **10**, if desired. In the preferred embodiment, the exemplary controller **12** should determine that the bung plug **34** in FIG. **38** has been secured to the drum **30** too far into the bung opening **32** and provide a signal that this drum **30** should be rejected and/or removed from the drum manipulation system **10**.

(168) Referring still to FIGS. **36-38**, any suitable configuration of components and techniques may be used to measure one or more bung plug insertion variable(s), use such data and direct any responsive actions. For example, during threadable engagement of the bung plug **34** with the drum **30**, one or more displacement sensor(s) **378** (e.g. linear encoder **380**) may be configured to monitor the amount of vertical movement, depth or distance that the bung plug **34** moves into the bung opening **32** during installation (e.g. on a continuing basis, in real-time), for any other purpose(s) or a combination thereof. For example, the displacement sensor **378** (e.g. FIGS. **45-48**) may be coupled (e.g. via bracket **382**) to the floating plate **254** (or other component). In some embodiments, the displacement sensor **378** (or other component(s)) may communicate data relating to vertical displacement measurements taken to the electronic controller **12** (or other component(s)) in real-time on a continuing basis for a determination of when rotation of the bung plug **34** should be stopped and to cause such stoppage and/or for any other purpose(s).

(169) For another example, the exemplary bung plug installer **60** may include one or more electronic, or other form, of torque sensors, torque transducers, torque meters or the like (e.g. one or more encoders **352** inside driver **204**, FIG. **35**) to measure the amount of torque applied to the bung plug **34** during installation (e.g. on a continuing basis, in real-time) and/or any related variable. For still a further example, the bung plug installer **60** may include one or more electronic,

or other forms, of counters (e.g. encoder **352** of driver **204**, FIG. **35**) to count or monitor the number of revolutions of rotation of the bung plug **34** (e.g. on a continuing basis, in real-time) as it is installed in the drum **30**. In the preferred embodiment, the driver **204** (e.g. servo-motor **350**) is capable of counting or monitoring the number of revolutions of rotation of the bung plug **34** as it is installed in the drum **30** in real-time on a continuing basis, the amount of torque applied to the bung plug **34** in real-time on a continuing basis during installation, one or more variations thereof or a combination thereof. For example, the servo-motor **350** may include one or more (e.g. internal) encoders **352** (FIG. **35**) and/or other component(s) configured to monitor, measure or determine one or more parameters that relate to the torque applied to the bung plug **34** and count or monitor the number of revolutions of rotation of the bung plug **34** as it is installed in the drum **30** during rotation thereof (e.g. on a continuing basis, in real-time). In some embodiments, this may entail the encoder **352** measuring resistance on the shaft **208** of the servo-motor **350** (e.g. FIG. **36**), demand placed upon the motor **350**, other variable(s) or a combination thereof. If desired, the encoder **352** may communicate its measurements and/or related data to the electronic controller **12** (or other component(s)) in real-time for a determination of when rotation of the bung plug **34** should be stopped and to cause such stoppage (e.g. when a desired torque had been applied to the bung plug **34**) and/or for any other purpose(s). In other embodiments, the driver **204** may be an air motor or one or more other components configured to assist in measuring or providing one or more bung plug insertion variables.

(170) When one or more bung plug insertion variable(s) or variations thereof fails to satisfy a desired value, the exemplary controller **12** (or other component(s)) may be configured to direct one or more responsive actions, such as reporting a fault event, providing a fault signal, initiating an alarm, rejecting or tagging the drum **30**, directing the bung plug installer **60** to uninstall or remove the bung plug **34**, ceasing operation of the bung plug installer **60**, drum manipulation system **10** or one or more components thereof, other action(s) or a combination thereof. In the preferred embodiment, for example, if the torque applied to the bung plug **34** exceeds a particular value (or range) prior to the occurrence of a particular number of revolutions of rotation of, or amount of downward vertical displacement, of the bung plug **34**, the controller **12** may be configured to unscrew bung plug **34** and repeat the installation process as described above up to two (or fewer or more) additional times before rejecting the drum **30**. For another example, if the torque applied to the bung plug **34** exceeds a particular value (or range) after the occurrence of a particular number of revolutions of rotation of, or amount of downward vertical displacement, the controller **12** may be configured to stop rotation of the bung plug **34** and identify or reject the drum **30** for removal from the drum manipulation system **10** (e.g. because the bung plug **34** is damaged or something else is wrong).

(171) Referring now to FIGS. **1A** & **39**, in the preferred embodiment, after the bung plug **34** is secured to the drum **30** (or earlier as desired), the exemplary bung plug engagement tool **184** of the bung plug installer **60** may be disengaged from the bung plug **34** and the bung plug installer **60** shuttled, or moved, away from the drum **30**, such as described above. For example, the bung plug installer **60** may be shuttled back to its home position or an intermediate position (e.g. to make room for the dust cap crimper **98** to crimp a dust cap over the bung plug **34**). In embodiments involving the use of suction to engage the bung plug **34**, the vacuum may be turned off or suction otherwise released (and air pressure reversed, if desired) at any desired time during or after the bung plug **34** is engaged with the drum **30**.

(172) Preferred embodiments of the present disclosure thus offer advantages over the prior art and are well adapted to carry out one or more of the objects of this disclosure. However, the present invention does not require each of the components and acts described above and is in no way limited to the above-described embodiments or methods of operation. Any one or more of the above components, features and processes may be employed in any suitable configuration without inclusion of other such components, features and processes. Moreover, the present invention

includes additional features, capabilities, functions, methods, uses and applications that have not been specifically addressed herein but are, or will become, apparent from the description herein, the appended drawings and/or claims.

(173) The methods described above or claimed herein and any other methods which may fall within the scope of the appended claims can be performed in any desired or suitable order and are not necessarily limited to any sequence described herein or as may be listed in the appended claims. Further, the methods of the present disclosure do not necessarily require use of the particular embodiments shown and described herein, but are equally applicable with any other suitable structure, form and configuration of components.

(174) While exemplary embodiments have been shown and described, many variations, modifications and/or changes of the system, apparatus and methods of the present disclosure, such as in the components, details of construction and operation, arrangement of parts and/or methods of use, are possible, contemplated by the patent applicant(s) hereof, within the scope of any appended claims, and may be made and used by one of ordinary skill in the art without departing from the spirit, teachings and scope of this disclosure and any appended claims. Thus, all matter herein set forth or shown in the accompanying drawings should be interpreted as illustrative, and the scope of the disclosure and any appended claims should not be limited to the embodiments described and shown herein.

Claims

1. Apparatus for rotating a container in a container manipulation system, the container being delivered to the container manipulation system on a conveyor, the conveyor having first and second sides and front and rear ends, the apparatus comprising: first and second carriages being positioned proximate to opposite sides of the conveyor, respectively, whereby the first carriage is closer to the first side of the conveyor and the second carriage is closer to the second side of the conveyor, each carriage carrying respective forward and rear clamp arms, the rear clamp arm of each carriage being closer to the rear end of the conveyor than the respective associated front clamp arm, and each clamp arm being selectively independently moveable toward and away from the container on the conveyor and carrying at least first and second vertically spaced-apart wheels, whereby the wheels on the respective rear clamp arms are rearward of the wheels on the corresponding associated front clamp arms relative to the conveyor, wherein all of the wheels are engageable with the container on the conveyor, at least two of the wheels on the first carriage are power-driven and selectively rotatable and the remaining wheels on the first and second carriages are free-spinning.
